Chapter 4
Aircraft Drawings

Introduction

The exchange of ideas is essential to everyone, regardless of their vocation or position. This exchange is usually carried on by oral or written word; but under some conditions, the use of these alone is impractical. The aviation industry discovered that it could not depend entirely upon written or spoken words for the exchange of ideas, because misunderstanding and misinterpretation arose frequently. A written description of an object can be changed in meaning just by misplacing a comma, and the meaning of an oral description can be completely changed by using a wrong word. To avoid these possible errors, drawings are used to describe objects. For this reason, drawing is the draftsman's language.

Drawing, in the aviation industry, is a method of conveying ideas concerning the construction or assembly of objects. This is done with the help of lines, notes, abbreviations, and symbols. It is important that the aviation mechanic who is to make or assemble the object understands the meaning of the different lines, notes, abbreviations, and symbols that are used in a drawing. (See the "Lines and Their Meanings" section of this chapter.)

Computer Graphics

From the early days of aviation, development of aircraft, aircraft engines, and other components relied heavily on aircraft drawings. For most of the 20th century, drawings were created on a drawing "board" with pen or pencil and paper. With the introduction and advancement of computers in the later decades of the 20th century, the way drawings are created changed dramatically. Computers were used not only to create drawings, but they were being used to show items in "virtual reality," from any possible viewing angle. Further development of computer software programs allowed for assembling of separately created parts to check for proper fit and possible interferences. Additionally, with nearly instantaneous information sharing capability through computer networking and the Internet, it became much easier for designers to share their work with other designers and manufacturers virtually anytime, anywhere in the world. Using new computer-controlled manufacturing techniques, it became possible to design a part and have it precisely manufactured without ever having shown it on paper. New terms and acronyms became commonplace. The more common of these terms are:

- Computer Graphics—drawing with the use of a computer
- Computer Aided Design (CAD)—where a computer is used in the design of a part or product
- Computer Aided Design Drafting (CADD)—where a computer is used in the design and drafting process
- Computer Aided Manufacturing (CAM)—where a computer is used in the manufacturing of a part or product
- Computer Aided Engineering (CAE)—where a computer is used in the engineering of a part or product

As computer hardware and software continue to evolve, a greater amount of CAE is completed in less time, at lower cost. In addition to product design, some of the other uses of CAE are product analysis, assembly, simulations, and maintenance information. *[Figure 4-1]*

CATIA, ProEngineer, Solid Works, and Unigraphics are some of the more popular CAD software packages used for aircraft design and manufacturing. Most airframe manufacturers use CATIA software to design their aircraft. The complete aircraft is designed and assembled in the software package before it is manufactured. Drawings of all parts of the aircraft are available and can be accessed using the computer software. Drawings are no longer limited to 1, 2, or 3 views. Drawings from every angle can easily be accessed by using the computer model of the part or product. Technicians can access drawings and aircraft manuals on laptops or even mobile devices when performing maintenance on the shop floor.

Purpose & Function of Aircraft Drawings

Drawings and prints are the link between the engineers who design an aircraft and the workers who build, maintain, and repair it. A print may be a copy of a working drawing for an aircraft part or group of parts, or for a design of a system or group of systems. They are made by placing a tracing of the drawing over a sheet of chemically-treated paper and exposing it to a strong light for a short period of time. When the exposed paper is developed, it turns blue where the light has penetrated the transparent tracing. The inked lines of the tracing, having blocked out the light, show as white lines on a blue background. With other types of sensitized paper, prints may have a white background with colored lines or a colored background with white lines.

Drawings created using computers may be viewed on the computer monitor or printed out in "hard copy" by use of an

ink jet or laser printer. Larger drawings may be printed by use of a plotter or large format printer. Large printers can print drawings up to 42 inches high with widths up to 600 inches by use of continuous roll paper. *[Figure 4-2]*

Care & Use of Drawings

Drawings should be handled carefully as they are both expensive and valuable. Open drawings slowly and carefully to prevent tearing of the paper. When the drawing is open, smooth out the fold lines instead of bending them backward.

To protect drawings from damage, never spread them on the floor or lay them on a surface covered with tools or other objects that may make holes in the paper. Hands should be free of oil, grease, or other unclean matter that can soil or smudge the print.

Never make notes or marks on a print, as they may confuse others and lead to incorrect work. Only authorized individuals are permitted to make notes or changes on prints, and they must sign and date any changes they make.

When finished with a drawing, fold and return it to its proper place. Prints are folded originally in an appropriate size for filing. Care should be taken so that the original folds are always used.

Types of Drawings

Drawings must give information such as size and shape of the object and all its parts, specifications for material to be used, how the material is to be finished, how the parts are to be assembled, and any other information essential to making and assembling the object. Drawings may be divided into three classes: detail, assembly, and installation.

Detail Drawing

A detail drawing is a description of a single part, describing bylines, notes, and symbols the specifications for size, shape, material, and methods of manufacture to be used in making the part. Detail drawings are usually rather simple. When single parts are small, several detail drawings may be shown on the same sheet or print. *[Figure 4-3]*

Assembly Drawing

An assembly drawing is a description of an object made up of two or more parts. *[Figure 4-4]* It describes the object's size and shape. Its primary purpose is to show the relationship of the various parts. An assembly drawing is usually more complex than a detail drawing and is often accompanied by detail drawings of various parts.



Figure 4-1. *Computer graphics work station.*

Installation Drawing

An installation drawing is one that includes all necessary information for a part or an assembly in the final installed position in the aircraft. It shows the dimensions necessary for the location of specific parts with relation to the other parts and reference dimensions that are helpful in later work in the shop. *[Figure 4-5]*

Sectional View Drawings

A section or sectional view is obtained by cutting away part of an object to show the shape and construction at the cutting plane. The part or parts cut away are shown by using section (crosshatching) lines. Types of sections are described in the following paragraphs.

Full Section

A full section view is used when the interior construction or hidden features of an object cannot be shown clearly by exterior views. For example, *Figure 4-6* is a sectional view of a cable connector and shows the internal construction of the connector.

Half Section

In a half section, the cutting plane extends only halfway across the object, leaving the other half of the object as an exterior view. Half sections are used with symmetrical objects to show both the interior and exterior. *Figure 4-7* is a half sectional view of a Capstan servo.

Revolved Section

A revolved section drawn directly on the exterior view shows the shape of the cross section of a part, such as the spoke of a wheel. An example of a revolved section is shown in *Figure 4-8*.



Figure 4-2. *Large format printer.*

Removed Section

A removed section illustrates parts of an object. It is drawn like revolved sections, except it is placed at one side and often drawn to a larger scale than the view indicated to bring out pertinent details.

Figure 4-9 is an illustration of removed sections. Section A-A shows the cross-sectional shape of the object at cutting plane line A-A. Section B-B shows the cross-sectional shape at cutting plane line B-B. These sectional views are drawn to the same scale as the principal view.

Title Blocks

Every print must have some means of identification. This is provided by a title block. *[Figure 4-4A]* The title block consists of a drawing number and certain other data concerning the drawing and the object it represents. This information is grouped in a prominent place on the print, usually in the lower right-hand corner. Sometimes the title block is in the form of a strip extending almost the entire distance across the bottom of the sheet.

Although title blocks do not follow a standard form as far as layout is concerned, all of them present essentially the following information:

1. A drawing number to identify the print for filing purposes and to prevent confusing it with any other print.
2. The name of the part or assembly
3. The drawing scale
4. The date
5. The name of the firm
6. The name of the draftsmen, the checker, and the person approving the drawing

Drawing or Print Numbers

All prints are identified by a number that appears in a number block in the lower right corner of the title block. It may also be shown in other places—such as near the top border line, in the upper right corner, or on the reverse side of the print at both ends—so that the number shows when the print is folded or rolled. The purpose of the number is quick identification of a print. If a print has more than one sheet and each sheet has the same number, this information is included in the number block, indicating the sheet number and the number of sheets in the series. *[Figure 4-4B]*

Reference and Dash Numbers

Reference numbers that appear in the title block refer you to the numbers of other prints. When more than one detail is shown on a drawing, dash numbers are used. Both parts would have the same drawing number plus an individual number, such as 40267-1 and 40267-2.

In addition to appearing in the title block, dash numbers may appear on the face of the drawing near the parts they identify. Dash numbers are also used to identify right-hand and left-hand parts.

In aircraft, many parts on the left side are like the corresponding parts on the right side but in reverse. The left-hand part is always shown in the drawing. The right-hand part is called for in the title block. Above the title block a notation is found, such as: 470204-1LH shown; 470204-2RH opposite. Both parts carry the same number, but the part called for is distinguished by a dash number. Some prints have odd numbers for left-hand parts and even numbers for right-hand parts.

Universal Numbering System

The universal numbering system provides a means of identifying standard drawing sizes. In the universal numbering system, each drawing number consists of six or seven digits. The first digit is always 1, 2, 4, or 5 and indicates the size of the drawing. The number 1 indicates a drawing of 8½" × 11"; number 2 indicates an 11" × 17" drawing; number 4 represents a drawing of 17" × 22"; and 5 indicates a width of between 17 and 36 inches but on a continuous roll. Letters are also used (and becoming more prevalent) with the most common letters being A through E. The letter A is 8½" × 11", B is 11" × 17", C is 17" × 22", D is 22" × 34" and E is 34" × 44". There are additional letters, such as D1 at 24" × 36", E1 at 30" × 42" and additional sizes unique to even larger formats but generally reserved for inter-company operations.

The remaining digits identify the drawing. Many firms have modified this basic system to conform to their needs. The letter or number depicting the standard drawing size may be

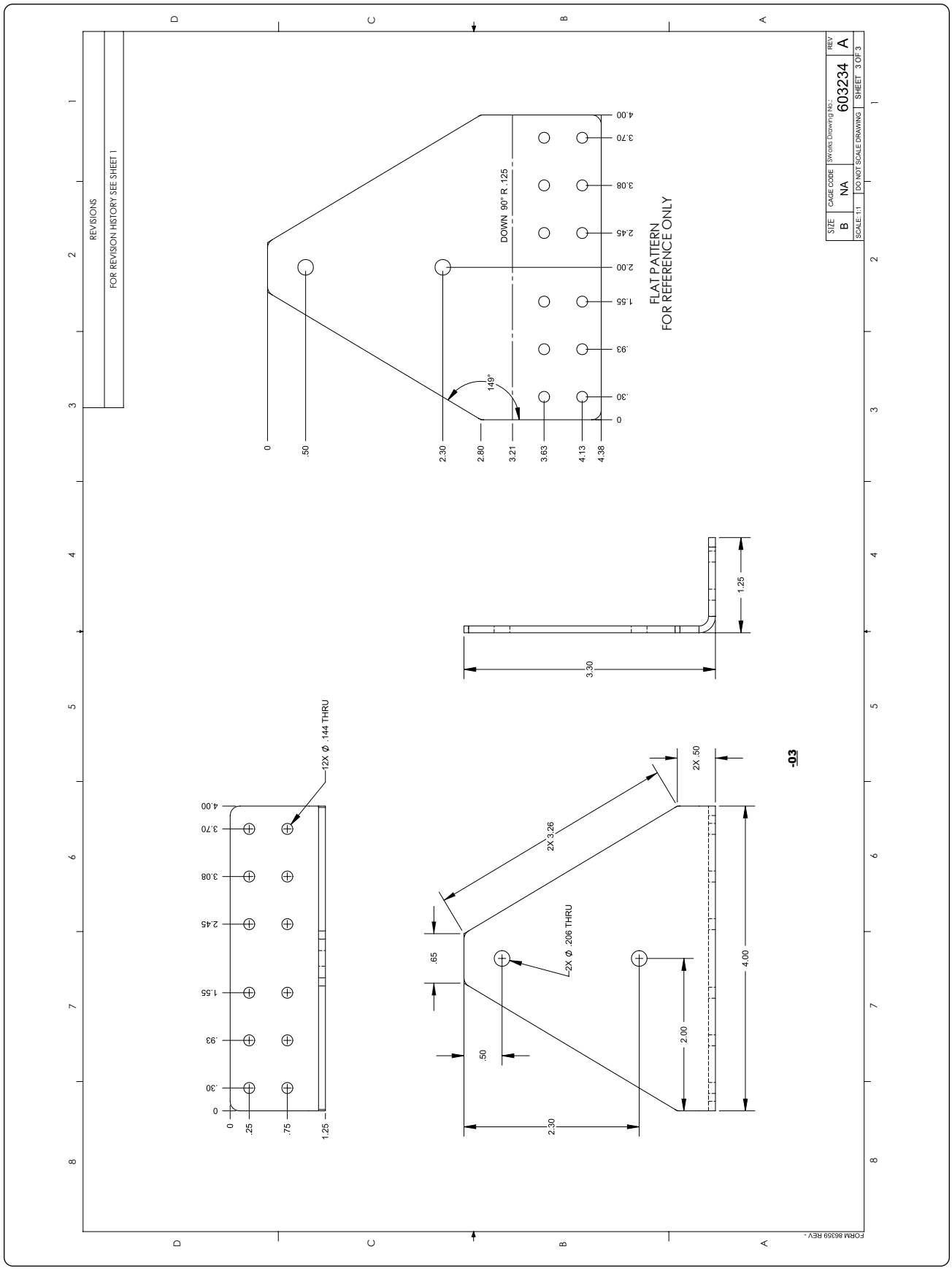


Figure 4-3. *Detail drawing.*

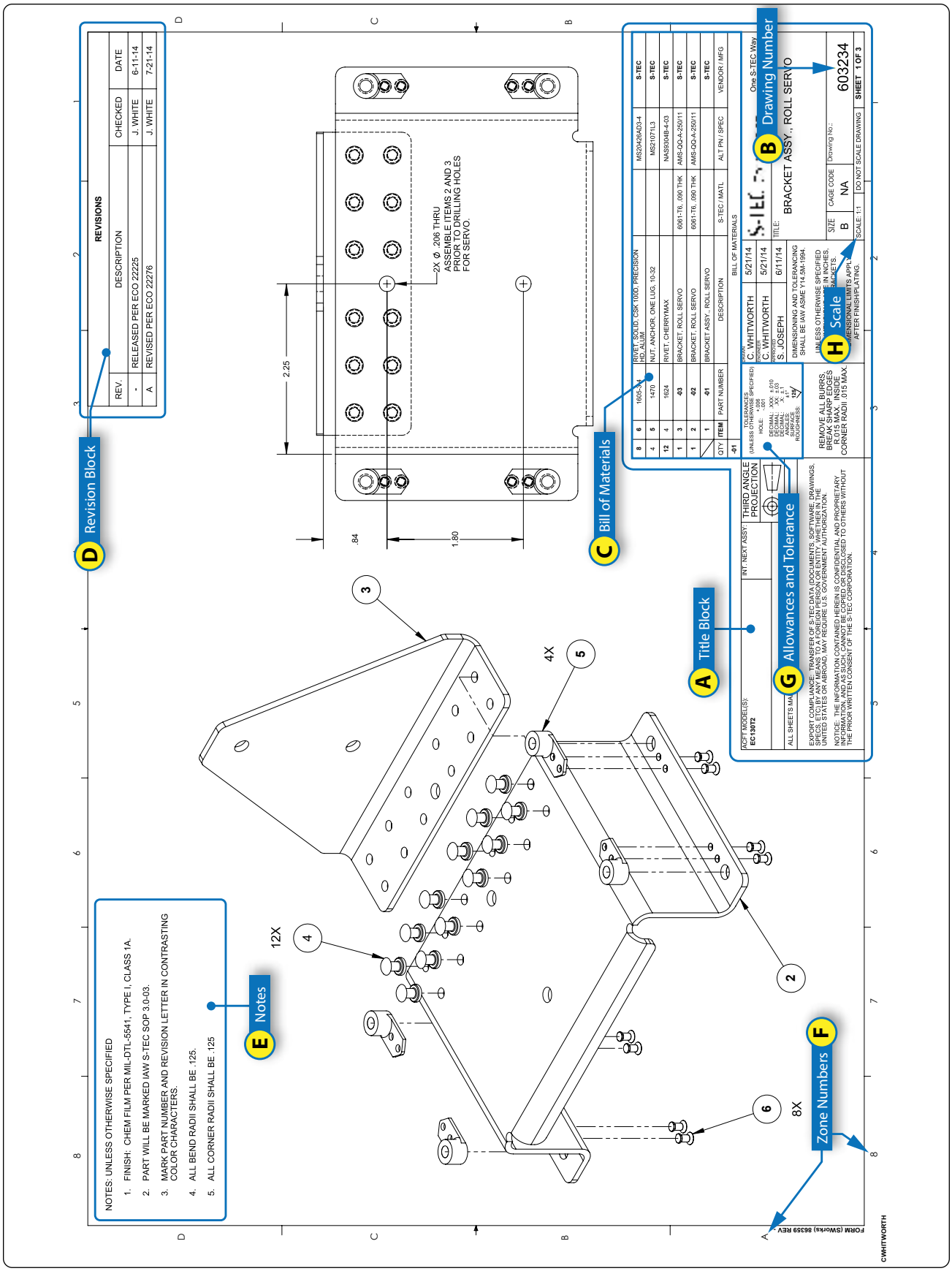


Figure 4-4. *Assembly drawing.*

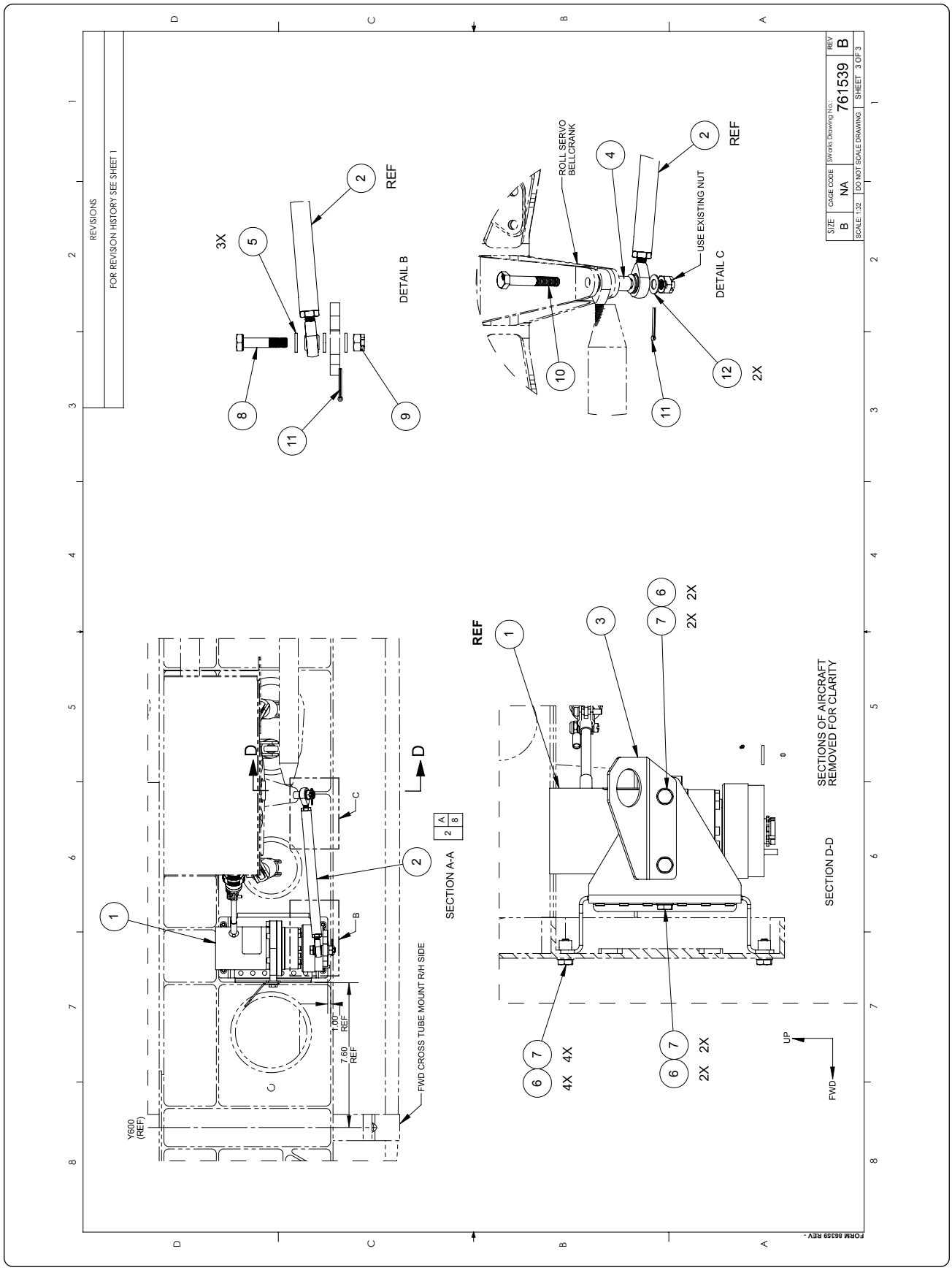


Figure 4-5. *Installation drawing.*

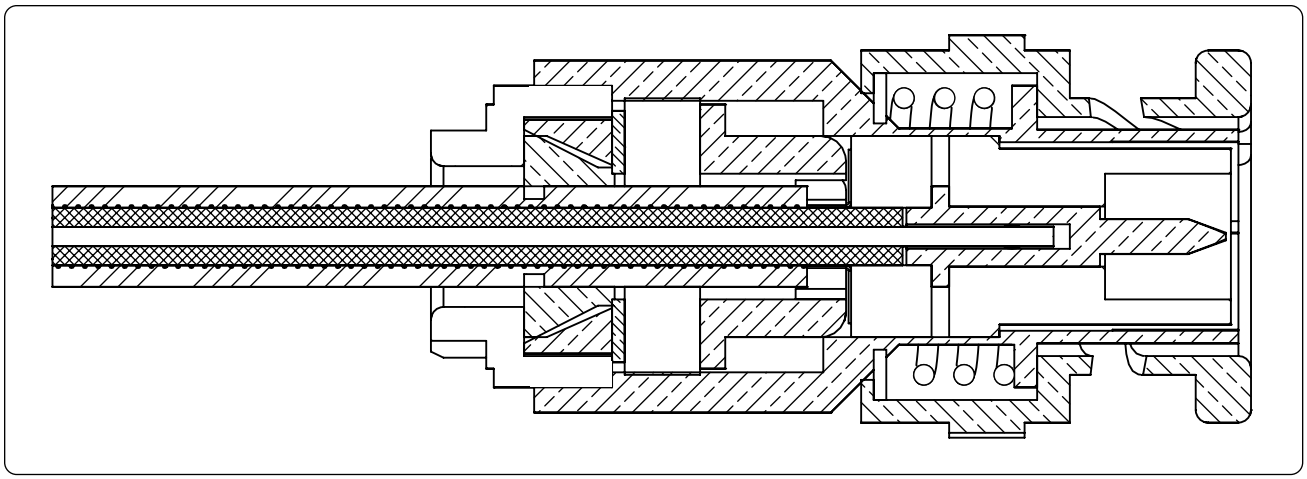


Figure 4-6. *Sectional view of a cable connector.*

prefixed to the number, separated from it by a dash. Other numbering systems provide a separate box preceding the drawing number for the drawing size identifier. In another modification of this system, the part number of the depicted assembly is assigned as the drawing number.

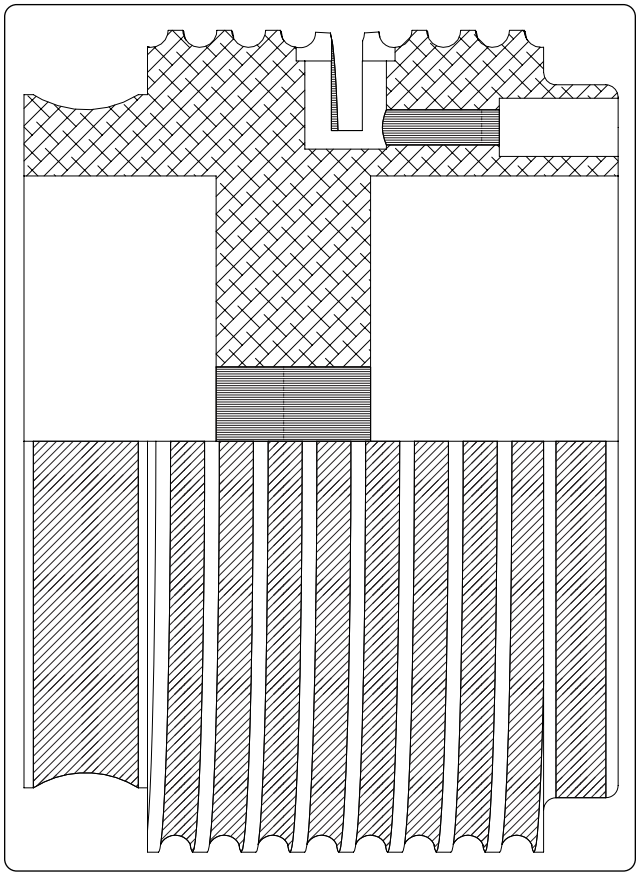


Figure 4-7. *Half section of a Capstan servo.*

Drawing Standards

Drawing standards cover such items as paper sizes, notes, numbering systems, geometric dimensions and tolerances, abbreviations, welding symbols, roughness symbols, and electrical symbols. These standards cover metric and inch measurements, as well as computer-drafting standards. Different standards for drawings are used in industry and some of the more common ones are published by the International Organization for Standardization (ISO) and the American National Standards Institute (ANSI).

Bill of Material

A list of the materials and parts necessary for the fabrication or assembly of a component or system is often included on the drawing. The list is usually in ruled columns that

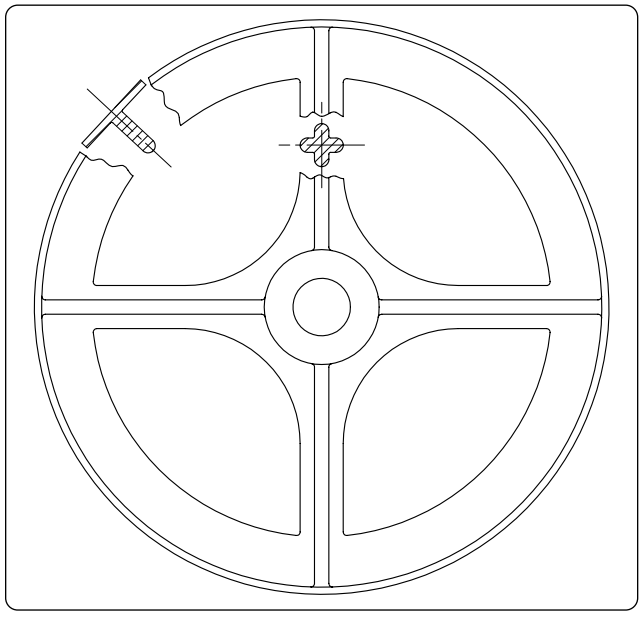


Figure 4-8. *Revolved sections.*

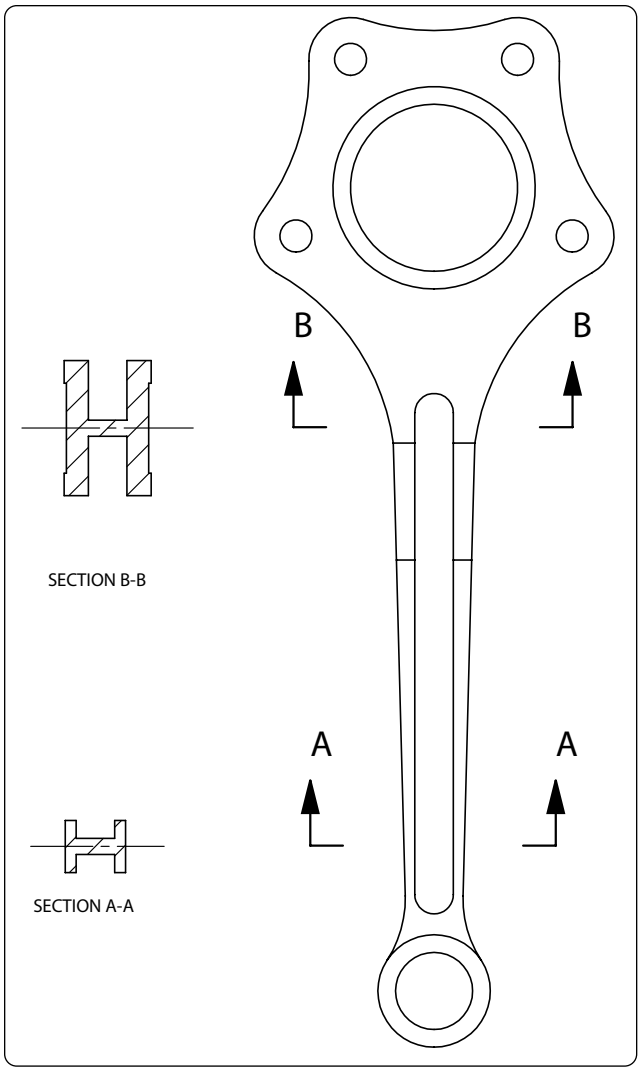


Figure 4-9. *Removed sections.*

provide the part number, name of the part, material the part is to be constructed of, the quantity required, and the source of the part or material. A typical bill of material is shown in *Figure 4-4C*. On drawings that do not have a bill of material, the data may be indicated directly on the drawing. On assembly drawings, each item is identified by a number in a circle or square. An arrow connecting the number with the item assists in locating it in the bill of material.

Other Drawing Data

Revision Block

Revisions to a drawing are necessitated by changes in dimensions, design, or materials. The changes are usually listed in ruled columns either adjacent to the title block or at one corner of the drawing. All changes to approved drawings must be carefully noted on all existing prints of the drawing.

When drawings contain such corrections, attention is directed to the changes by lettering or numbering them and listing those changes against the symbol in a revision block. *[Figure 4-4D]* The revision block contains the identification symbol, the date, the nature of the revision, the authority for the change, and the name of the draftsman who made the change.

To distinguish the corrected drawing from its previous version, many firms are including, as part of the title block, a space for entering the appropriate symbol to designate that the drawing has been changed or revised.

Notes

Notes are added to drawings for various reasons. Some of these notes refer to methods of attachment or construction. Others give alternatives, so that the drawing can be used for different styles of the same object. Still others list modifications that are available. Notes may be found alongside the item that they refer to. If the notes are lengthy, they may be placed elsewhere on the drawing and identified by letters or numbers. Notes are used only when the information cannot be conveyed in the conventional manner or when it is desirable to avoid crowding the drawing. *Figure 4-4E* illustrates one method of depicting notes.

When the note refers to a specific part, a light line with an arrowhead leads from the note to the part. If it applies to more than one part, the note is worded to eliminate ambiguity as to the parts it pertains to. If there are several notes, they are generally grouped together and numbered consecutively.

Zone Numbers

Zone numbers on drawings are like the numbers and letters printed on the borders of a map. They help locate a point. To find a point, mentally draw horizontal and vertical lines from the letters and numerals specified; the point where these lines intersect is the area sought. *Figure 4-4F* shows the zone numbers on a drawing.

Use the same method to locate parts, sections, and views on large drawings, particularly assembly drawings. Parts numbered in the title block can be located on the drawing by finding the numbers in squares along the lower border. Zone numbers read from right to left.

Station Numbers & Location Identification on Aircraft

A numbering system is used on large assemblies for aircraft to locate stations, such as fuselage frames. Fuselage station 185 indicates a location that is 185 inches from the datum of the aircraft. The measurement is usually taken from the nose or zero station, but in some instances, it may be taken from the firewall or some other point chosen by the manufacturer. Just as forward and aft locations on aircraft are made by

reference to the datum, locations left and right of the aircraft's longitudinal axis are made by reference to the buttock line and are called butt stations. Vertical locations on aircraft are made in reference to the waterline.

The same station numbering system is used for wing and stabilizer frames. The measurement is taken from the centerline or zero station of the aircraft. *Figure 4-10* shows use of the fuselage stations (FS), waterline locations (WL), and left and right buttock line locations (RBL and LBL).

Allowances & Tolerances

When a given dimension on a print shows an allowable variation, the plus (+) figure indicates the maximum, and the minus (−) figure the minimum allowable variation. The sum of the plus and minus allowance figures is called tolerance. *[Figure 4-4G]* For example, using 0.225 + 0.0025 − 0.0005, the plus and minus figures indicate the part is acceptable if it is not more than 0.0025 larger than the 0.225 given dimension, or not more than 0.0005 smaller than the 0.225 dimension. Tolerance in this example is 0.0030 (0.0025 max plus 0.0005 min).

If the plus and minus allowances are the same, you will find them presented as 0.225 ± 0.0025. The tolerance would then be 0.0050. Allowance can be indicated in either fractional or decimal form. When very accurate dimensions are necessary, decimal allowances are used. Fractional allowances are sufficient when precise tolerances are not required. Standard tolerances of –0.010 or −1/32 may be given in the title block of many drawings, to apply throughout the drawing.

Finish Marks

Finish marks are used to indicate the surface that must be machine finished. Such finished surfaces have a better appearance and allow a closer fit with adjoining parts. During the finishing process, the required limits and tolerances must be observed. Do not confuse machined finishes with those of paint, enamel, chromium plating, and similar coating.

Scale

Some drawings are made the same size as the drawn part; reflecting a scale of 1:1. Other scales may be used. However, when drawings are made on a computer, drawing sizes may be easily increased (zoom in) or decreased (zoom out). Some electronic printers have the same capability. Furthermore, when a 1:1 copy of a print is made, the copy size may differ slightly from that of the original. For accurate information, refer to the dimensions shown on the drawing. *[Figure 4-4H]*

Application

When shown near or in the title block, application may refer to a specific aircraft, assembly, sub-assembly or unique application. For example, in *Figure 4-4A* the title block indicates the bracket assembly is for a Roll Servo installation for an S-Tec Auto Pilot installation. If this drawing pertained to a B95 Aircraft equipped with an Aero-Tech air conditioning system and the bracket illustrated was unique to that installation, the title block would provide that application information. The title block may indicate Bracket Assy., Roll Servo, with Aero-Tech air conditioner (Model AT103-1) installed.

Methods of Illustration

Applied Geometry

Geometry is the branch of mathematics that deals with lines, angles, figures, and certain assumed properties in space. Applied geometry, as used in drawings, makes use of these properties to accurately and correctly represent objects graphically. In the past, draftsmen utilized a variety of instruments with various scales, shapes, and curves to make their drawings. Today, computer software graphics programs show drawings at nearly any scale, shape, and curve imaginable, outdating the need for additional instruments.

Several methods are used to illustrate objects graphically. The most common are orthographic projections, pictorial drawings, diagrams, and flowcharts.

Orthographic Projection Drawings

To show the exact size and shape of all the parts of complex objects, several views are necessary. This is the system used in orthographic projection.

In orthographic projection, there are six possible views of an object, because all objects have six sides—front, top, bottom, rear, right side, and left side. *Figure 4-11A* shows an object placed in a transparent box, hinged at the edges. The projections on the sides of the box are the views as seen looking straight at the object through each side. If the outlines of the object are drawn on each surface of the box, and the box is then opened *[Figure 4-11B]* to lay flat *[Figure 4-11C]*, the result is a six-view orthographic projection.

It is seldom necessary to show all six views to portray an object clearly; therefore, only those views necessary to illustrate the required characteristics of the object are drawn. One-, two-, and three-view drawings are the most common. Regardless of the number of views used, the arrangement is generally as shown in *Figure 4-11*, with the front view as principal view. If the right-side view is shown, it will be to the right of the front view. If the left-side view is shown, it will be to the left of the front view. The top and bottom views, if included, will be shown in their respective positions relative to the front view.

One-view drawings are commonly used for objects of

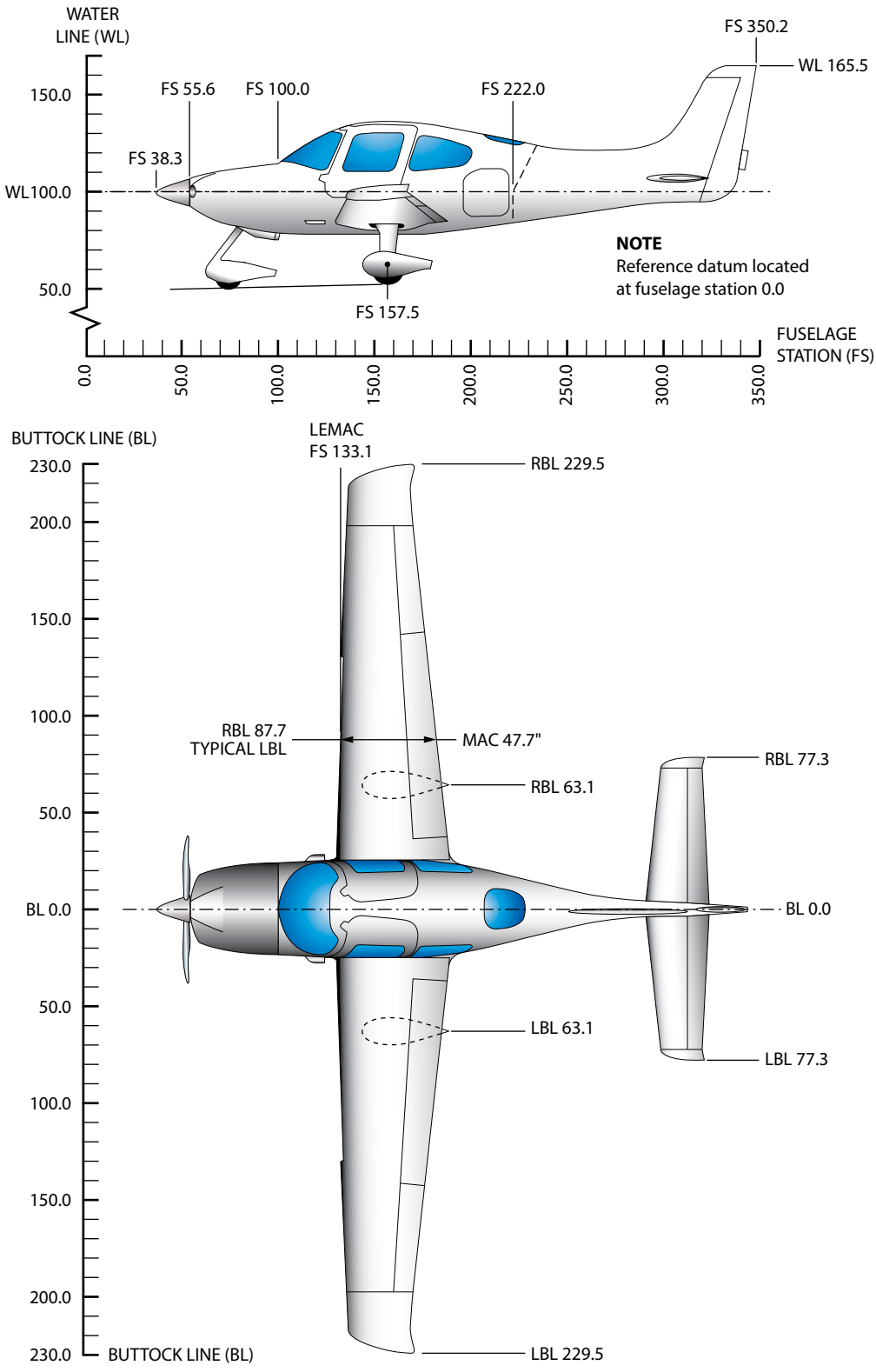


Figure 4-10. *Station numbers and location identification on aircraft.*

uniform thickness, such as gaskets, shims, and plates. A dimensional note gives the thickness as shown in *Figure 4-12*. One-view drawings are also commonly used for cylindrical, spherical, or square parts if all the necessary dimensions can be properly shown in one view. When space is limited and two views must be shown, symmetrical objects are often represented by half views, as illustrated in *Figure 4-13*.

Aircraft drawings seldom show more than two principal or complete views of an object. Instead, there will be usually one complete view and one or more detail views or sectional views.

Detail View

A detail view shows only a part of the object, but in greater detail and to a larger scale than the principal view. The part that is shown in detail elsewhere on the drawing is usually encircled by a heavy line on the principal view. *[Figure 4-14]* The principal view shows the complete object, while the detail view is an enlarged drawing of a portion of the object.

Pictorial Drawings

A pictorial drawing is like a photograph. *[Figure 4-15]* It shows an object as it appears to the eye, but it is not satisfactory for showing complex forms and shapes. Pictorial drawings are useful in showing the general appearance of an object and are used extensively with orthographic projection drawings. Pictorial drawings are used in Aircraft Maintenance Manuals (AMM), Structural Repair Manuals (SRM), and Illustrated Parts Catalogues (IPC). Four types of pictorial drawings used frequently by aircraft engineers and technicians are: perspective, isometric, oblique, and exploded view.

Perspective Drawings

A perspective view shows an object as it appears to an observer. *[Figure 4-16A]* It most closely resembles the way an object would look in a photograph. Because of perspective, some of the lines of an object are not parallel and therefore the actual angles and dimensions are not accurate.

Isometric Drawings

An isometric view uses a combination of the views of an orthographic projection and tilts the object forward so that portions of all three views can be seen in one view. *[Figure 4-16B]* This provides the observer with a three-dimensional view of the object. Unlike a perspective drawing where lines converge and dimensions are not true, lines in an isometric drawing are parallel and dimensioned as they are in an orthographic projection.

Oblique Drawings

An oblique view is like an isometric view, except for one distinct difference. In an oblique drawing, two of the three drawing axes are always at right angles to each other. *[Figure 4-16C]*

Exploded View Drawings

An exploded view drawing is a pictorial drawing of two or more parts that fit together as an assembly. The view shows the individual parts and their relative position to the other parts before they are assembled. *[Figure 4-17]*

Exploded view drawings are often used in IPCs that are used to order parts. The exploded view drawing has numbers and the numbers correspond to a list of part numbers. Exploded views are also used in Maintenance Instruction Manuals (MIM) for the assembly and repair of aircraft components. These drawings are often accompanied by notes that explain the assembly process.

Diagrams

A diagram may be defined as a graphic representation of an assembly or system, indicating the various parts and expressing the methods or principles of operation. There are many types of diagrams; however, those that the aviation mechanic is concerned with during the performance of their job may be grouped into four classes or types: installation, schematic, block, and wiring diagrams.

Installation Diagrams

Figure 4-18 is an example of an installation diagram. This is a diagram of the installation of the flight guidance control components of an aircraft. It identifies each of the components in the systems and shows their location in the aircraft. Each number (1, 2, 3, and 4) on the detail shows the location of the individual flight guidance system components within the flight deck of the aircraft. Installation diagrams are used extensively in aircraft maintenance and repair manuals, and are invaluable in identifying and locating components and understanding the operation of various systems.

Schematic Diagrams

Schematic diagrams do not indicate the location of the individual components in the aircraft nor do they show the actual size and shape of the components, but rather locate components with respect to each other within the system. Schematics show the principle of operation of an aircraft system and are often used for troubleshooting and training purposes.

Figure 4-19 illustrates a schematic diagram of an aircraft air conditioning system. High-speed bleed air from the engine is

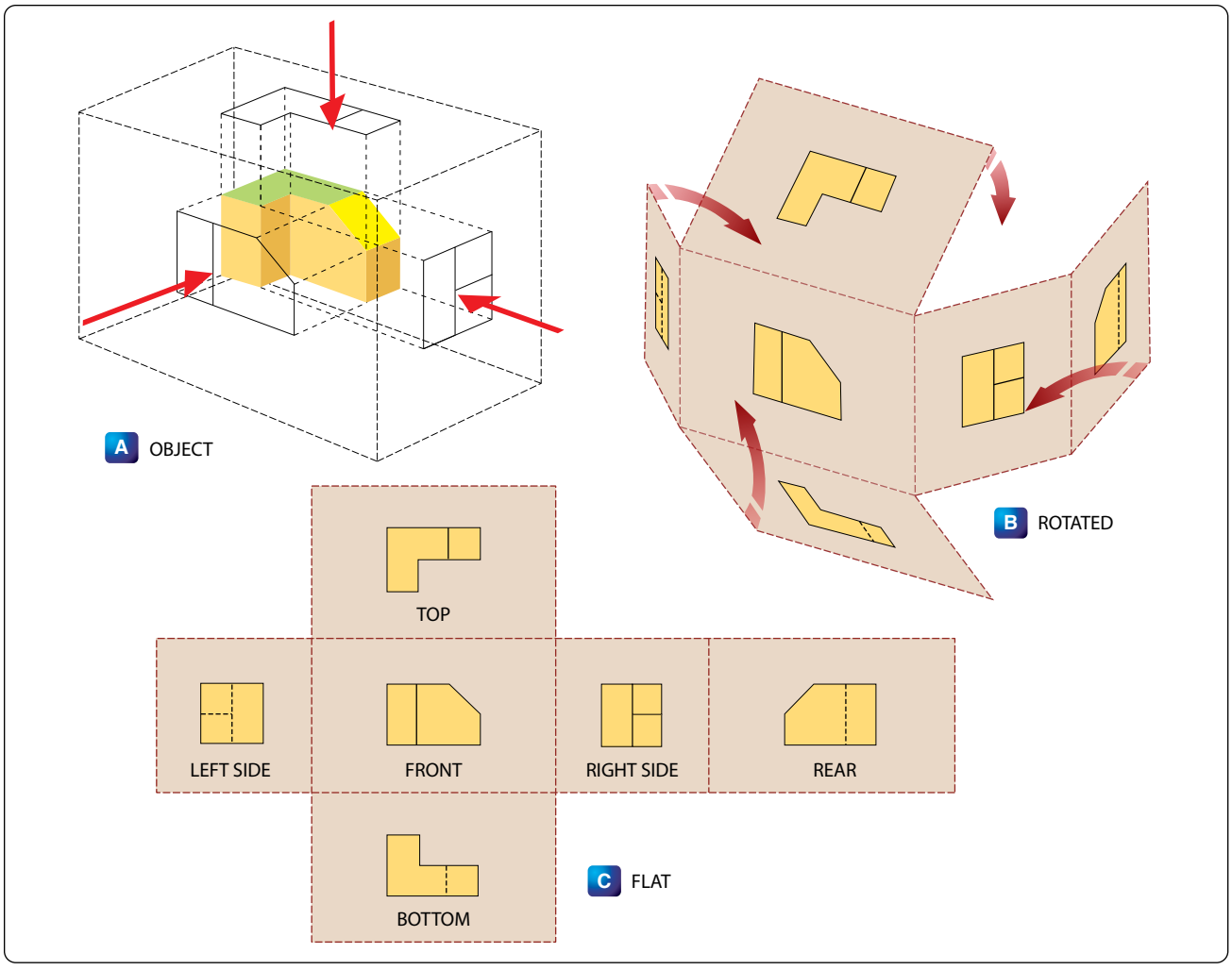


Figure 4-11. *Orthographic projection.*

combined with cold air in the mixing chamber and distributed via a manifold to various parts of the aircraft.

Note that each line is coded for ease of reading and tracing the flow. Each component is identified by name, and its location within the system can be ascertained by noting the lines that lead into and out of the unit. Schematic diagrams and installation diagrams are used extensively in aircraft manuals.

Block Diagrams

Block diagrams are used to show a simplified relationship of a

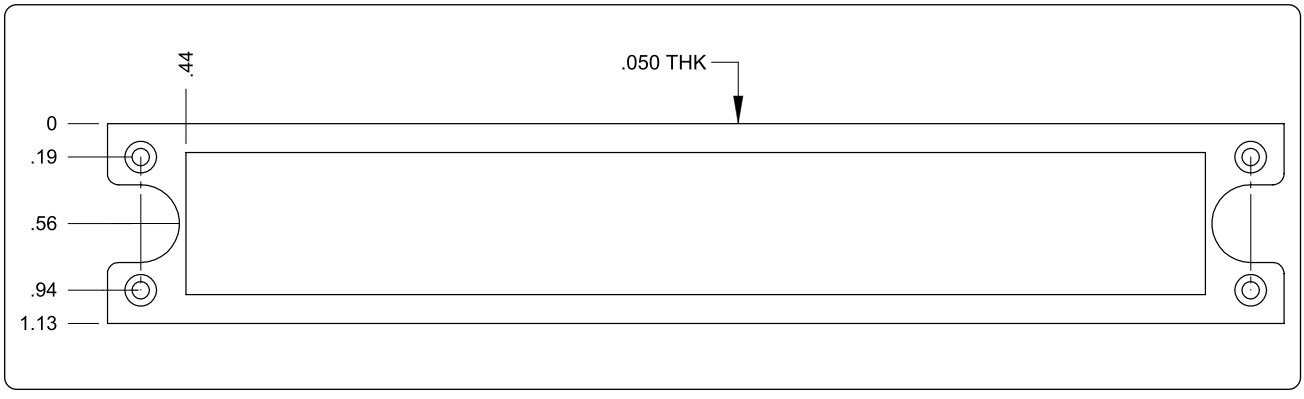


Figure 4-12. *One-view drawing.*

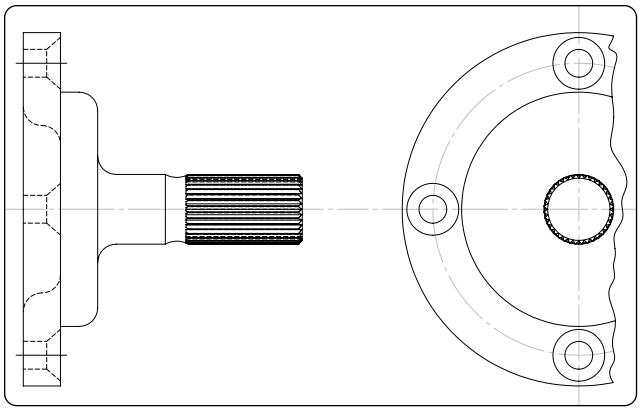


Figure 4-13. *Symmetrical object with exterior half view.*

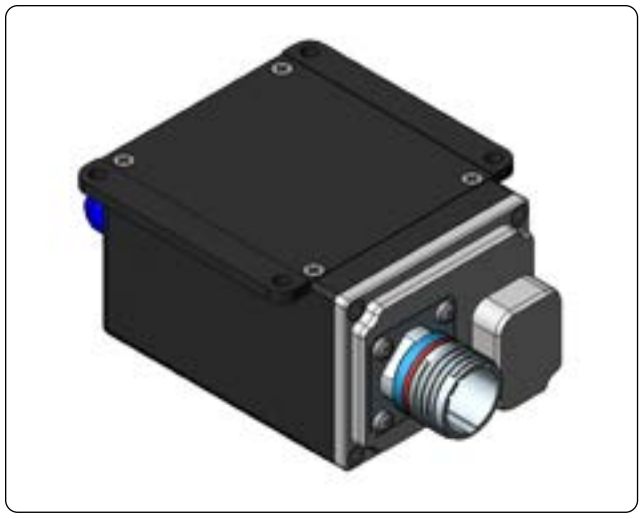


Figure 4-15. *Pictorial drawing.*

more complex system of components. *[Figure 4-20]* Individual components are drawn as a rectangle (block) with lines connecting it to other components (blocks) that it interfaces with during operation.

Wiring Diagrams

Wiring diagrams show the electrical wiring and circuitry, coded for identification, of all the electrical appliances and devices used on aircraft. *[Figure 4-21]* These diagrams, even for relatively simple circuits, can be quite complicated. For technicians involved with electrical repairs and installations, a thorough knowledge of wiring diagrams and electrical schematics is essential.

Flowcharts

Flowcharts are used to illustrate a sequence or flow of events. There are two types of flow charts most frequently

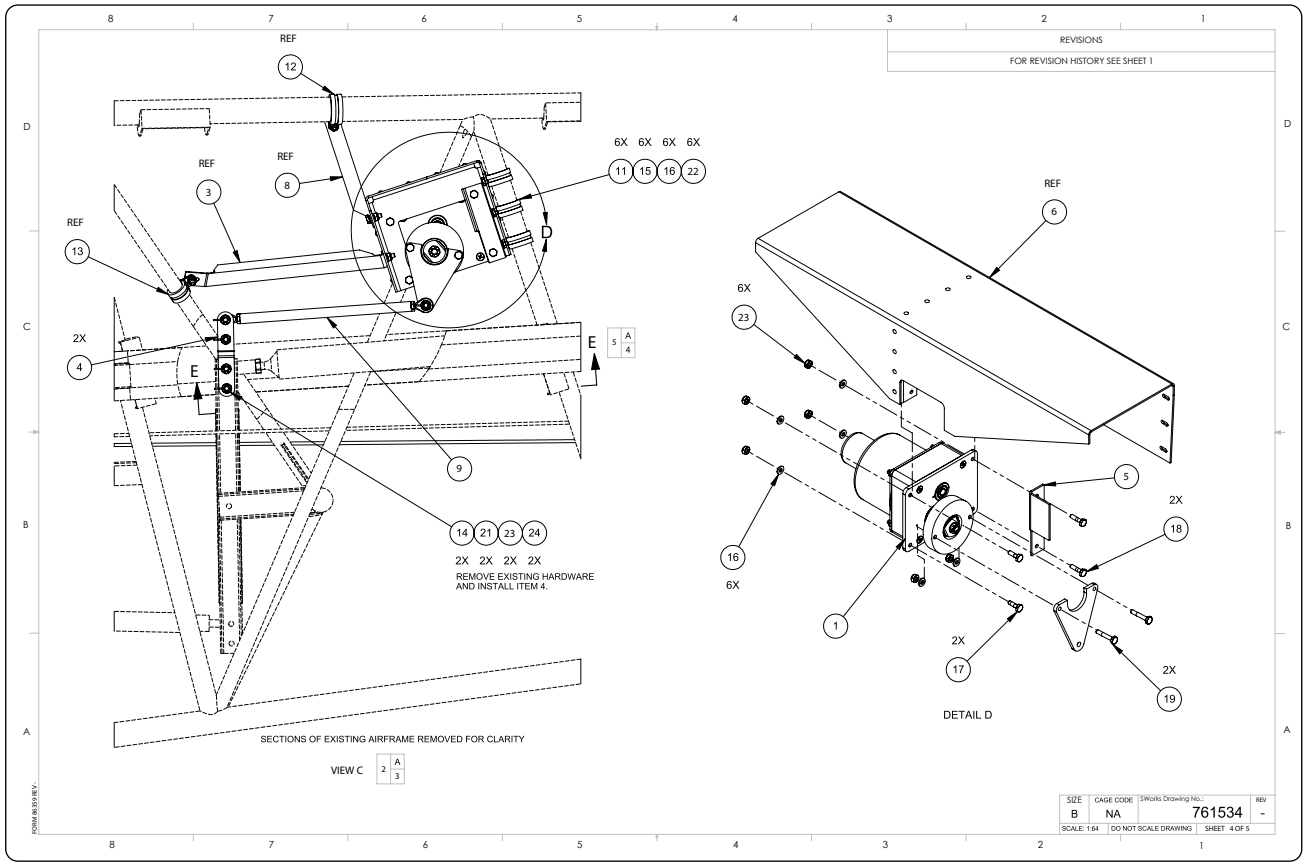


Figure 4-14. *Detail view.*

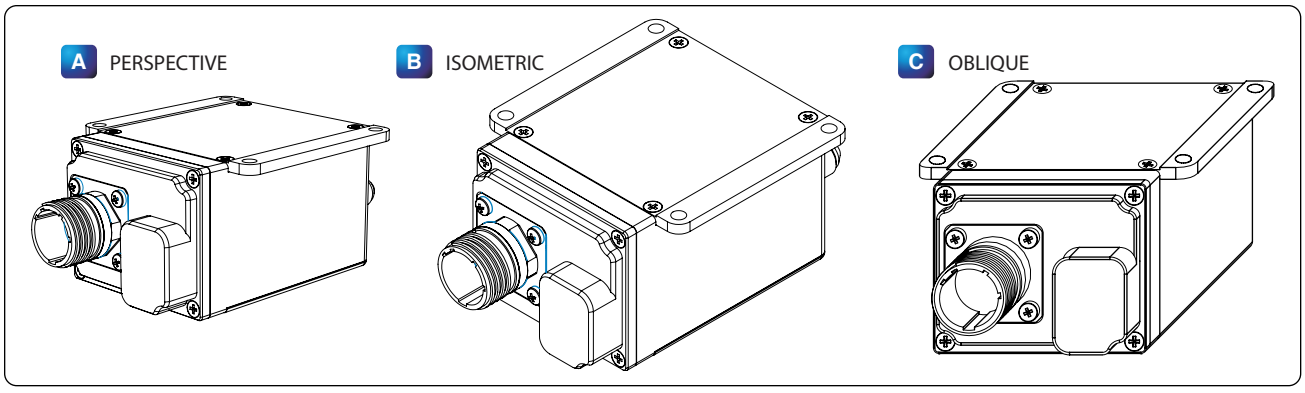
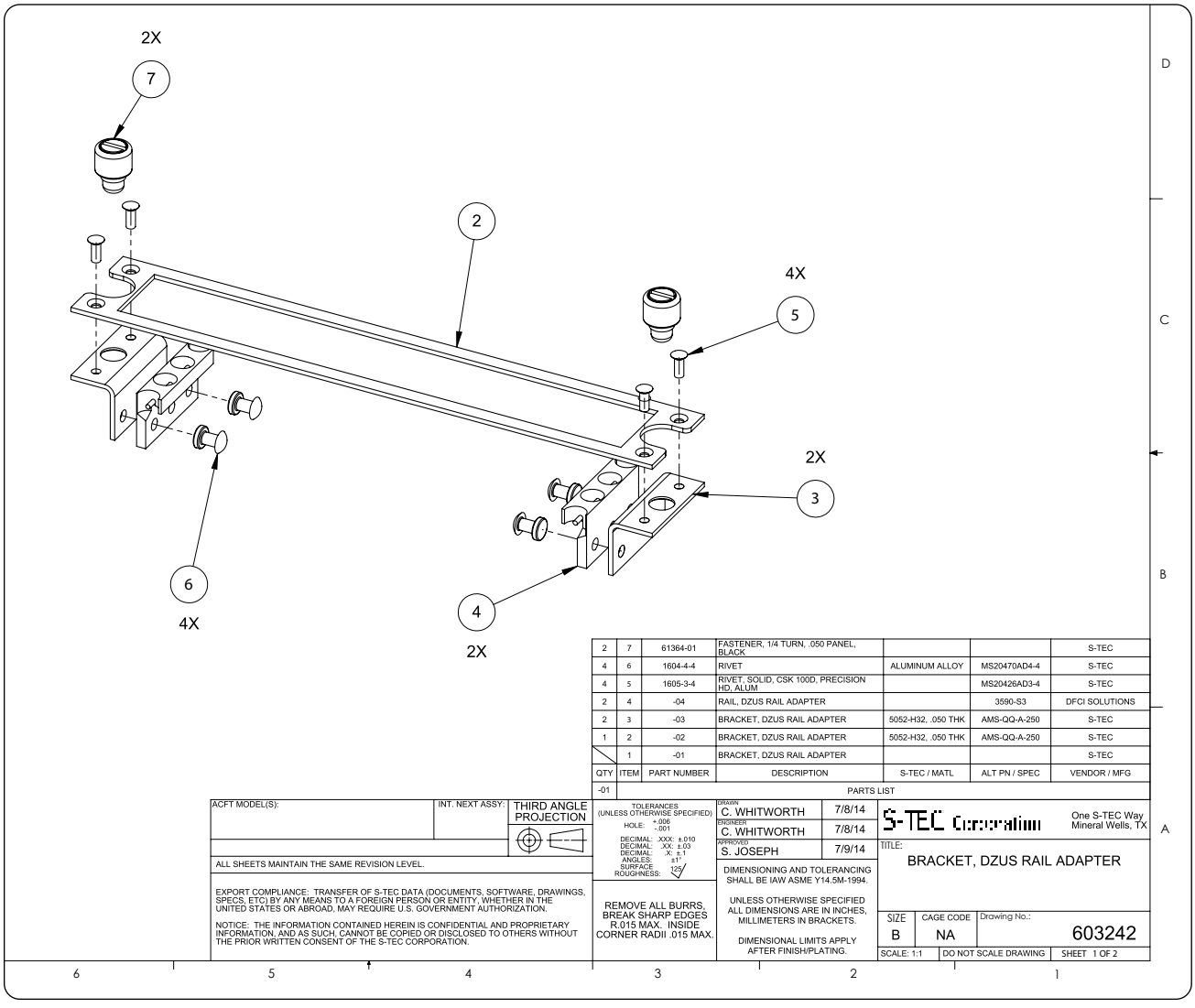


Figure 4-16. *(A) Perspective, (B) isometric, and (C) oblique drawings.*

used in the aviation industry: troubleshooting flowcharts and logic flowcharts.

Troubleshooting Flowchart

Troubleshooting flowcharts are frequently used for the detection of faulty components. They often consist of a series of yes or no questions. If the answer to a question



QTY	ITEM	PART NUMBER	DESCRIPTION	S-TEC / MATL	ALT PN / SPEC	VENDOR / MFG
-01			PARTS LIST			
2	7	61364-01	FASTENER, 1/4 TURN, .050 PANEL, BLACK			S-TEC
4	6	1604-4-4	RIVET	ALUMINUM ALLOY	MS20470AD4-4	S-TEC
4	5	1605-3-4	RIVET, SOLID, CSK 100D, PRECISION HD, ALUM		MS20426AD3-4	S-TEC
2	4	-04	RAIL, DZUS RAIL ADAPTER		3590-S3	DFCI SOLUTIONS
2	3	-03	BRACKET, DZUS RAIL ADAPTER	5052-H32, .050 THK	AMS-QQ-A-250	S-TEC
1	2	-02	BRACKET, DZUS RAIL ADAPTER	5052-H32, .050 THK	AMS-QQ-A-250	S-TEC
	1	-01	BRACKET, DZUS RAIL ADAPTER			S-TEC

ACFT MODEL(S): | INT. NEXT ASSY: | THIRD ANGLE PROJECTION

ALL SHEETS MAINTAIN THE SAME REVISION LEVEL.

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TOLERANCES (UNLESS OTHERWISE SPECIFIED)
HOLE: +.006 -.001
DECIMAL: XXX: ±.010
DECIMAL: XX: ±.03
DECIMAL: X: ± 1
ANGLES: ±1°
SURFACE ROUGHNESS: 125

REMOVE ALL BURRS, BREAK SHARP EDGES R.015 MAX. INSIDE CORNER RADII .015 MAX.

DRAWN: C. WHITWORTH 7/8/14
ENGINEER: C. WHITWORTH 7/8/14
APPROVED: S. JOSEPH 7/9/14

DIMENSIONING AND TOLERANCING SHALL BE IAW ASME Y14.5M-1994.

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES, MILLIMETERS IN BRACKETS.

DIMENSIONAL LIMITS APPLY AFTER FINISH/PLATING.

S-TEC Corporation
One S-TEC Way
Mineral Wells, TX

TITLE: BRACKET, DZUS RAIL ADAPTER

SIZE	CAGE CODE	Drawing No.:
B	NA	603242

SCALE: 1:1 | DO NOT SCALE DRAWING | SHEET 1 OF 2

Figure 4-17. *Exploded view drawing.*

is yes, one course of action is followed. If the answer is no, a different course of action is followed. In this simple manner, a logical solution to a problem may be achieved. *Figure 4-22* shows a flow chart to determine the repair options for a composite structure.

Logic Flowchart

Another type of flowchart, developed specifically for analysis of digitally-controlled components and systems, is the logic flowchart. *[Figure 4-23]* A logic flowchart uses standardized symbols to indicate specific types of logic gates and their relationship to other digital devices in a system. Since digital systems make use of binary mathematics consisting of 1s and 0s, voltage or no voltage, a light pulse or no light pulse, and so forth, logic flowcharts consist of individual components that take an input and provide an output that is either the same as the input or opposite. By analyzing the input or multiple inputs, it is possible to determine the digital output or outputs.

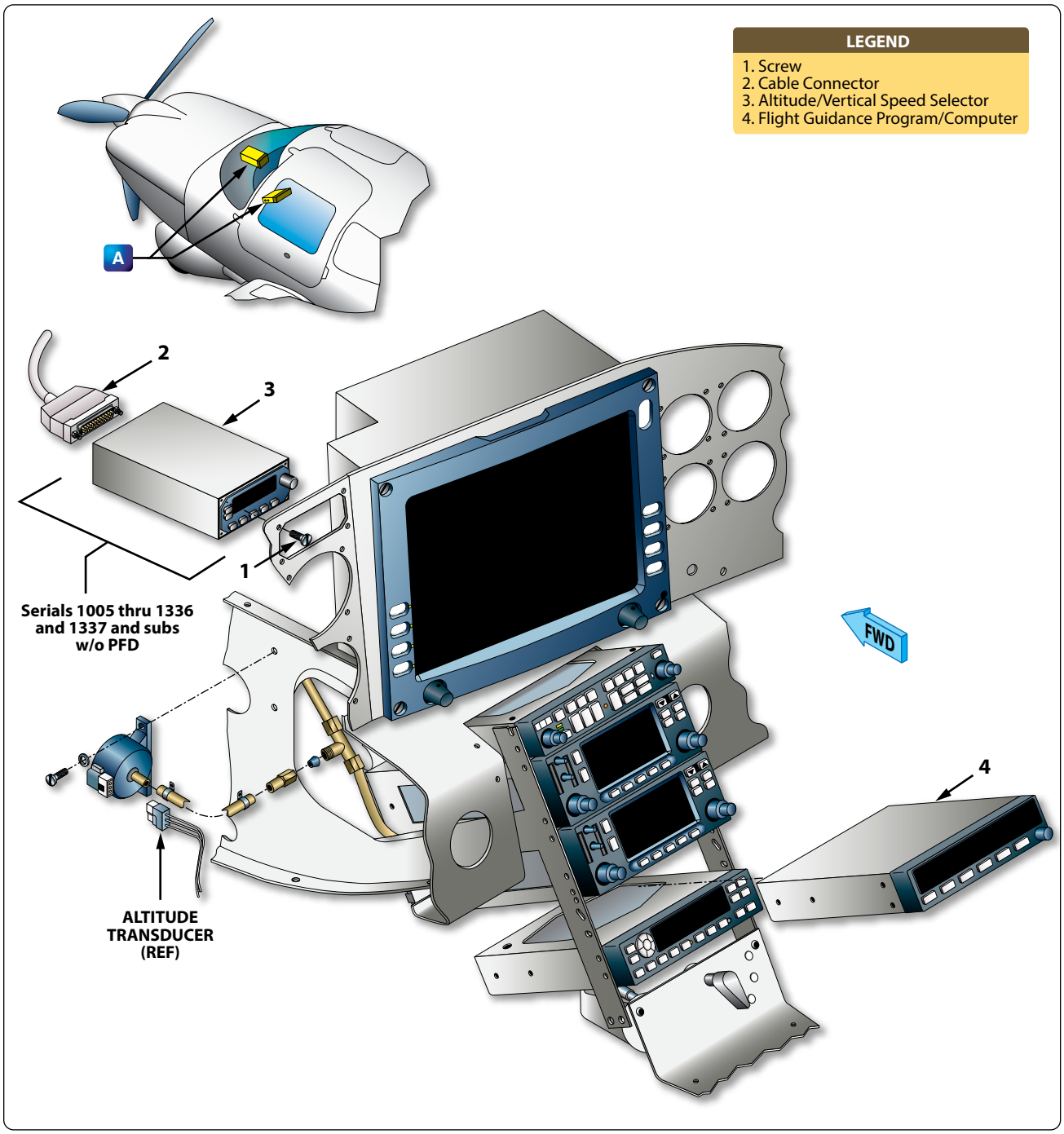


Figure 4-18. *Example of an installation diagram (flight guidance components).*

Lines and Their Meanings

Every drawing is composed of lines. Lines mark the boundaries, edges, and intersection of surfaces. Lines are used to show dimensions and hidden surfaces and to indicate centers. Obviously, if the same kind of line is used to show these variations, a drawing becomes a meaningless collection of lines. For this reason, various kinds of standardized lines are used on aircraft drawings. *[Figure 4-24]* Examples of correct line uses are shown in *Figure 4-25*.

Most drawings use three widths, or intensities, of lines: thin, medium, or thick. These lines may vary somewhat on different drawings, but there is a noticeable difference between a thin and a thick line, with the width of the medium line somewhere between the two.

Centerlines

Centerlines are made up of alternate long and short dashes. They indicate the center of an object or part of an object. Where centerlines cross, the short dashes intersect

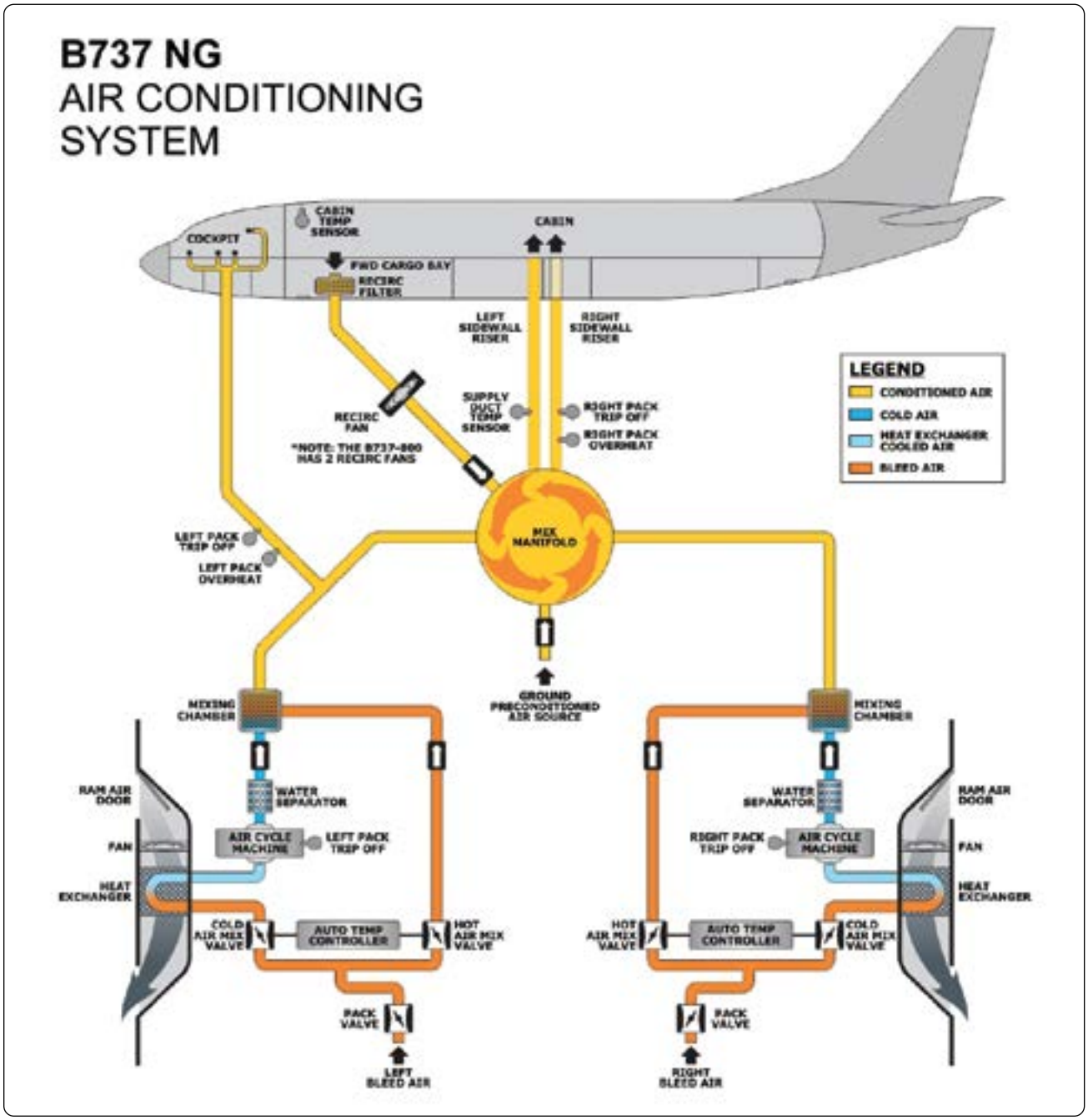


Figure 4-19. *Schematic diagram of an air conditioning system for a B737 NG.*

symmetrically. In the case of very small circles, the centerlines may be shown unbroken.

Dimension Lines

A dimension line is a light solid line, broken at the midpoint for insertion of measurement indications, and having opposite pointing arrowheads at each end to show origin and termination of a measurement. They are generally parallel to the line that the dimension is given for, placed outside the outline of the object, and between views if more than one view is shown.

All dimensions and lettering are placed so that they read from left to right. The dimension of an angle is indicated by placing the degree of the angle in its arc. The dimensions of circular parts are always given in terms of the diameter of the circle and are usually marked with the letter D or the abbreviation DIA following the dimension. The dimension of an arc is given in terms of its radius and is marked with the letter R following the dimension. Parallel dimensions are placed so that the longest dimension is farthest from the outline and the shortest dimension is closest to the outline of the object. On a drawing showing several views, the dimensions are placed upon each view to show its details to the best advantage.

In dimensioning distances between holes in an object, dimensions are usually given from center to center rather than from outside to outside of the holes. When several holes of various sizes are shown, the desired diameters are given on a leader followed by notes indicating the machining operations for each hole. If a part is to have three holes of equal size, equally spaced, this information is explicitly stated. For precision work, sizes are given in decimals. Diameters and depths are given for counterbored holes. For countersunk holes, the angle of countersinking and the diameters are given. *[Figure 4-26]*

The dimensions given for tolerances signify the amount of clearance allowable between moving parts. A positive allowance is indicated for a part that is to slide or revolve upon another part. A negative allowance is one given for a force fit. Whenever possible, the tolerance and allowances for desired fits conform to those set up in the American Standard for Tolerances, Allowances, and Gauges for Metal Fits. The classes of fits specified in the standard may be indicated on assembly drawings.

Extension Lines

Extensions are used to extend the line showing the side or edge of a figure for placing a dimension to that side or edge. They are very narrow and have a short break where they extend from the object and extend a short distance past the arrow of the dimensioning line.

Sectioning Lines

Sectioning lines indicate the exposed surfaces of an object in a sectional view. They are generally thin full lines, but may vary with the kind of material shown in section.

Phantom Lines

Phantom lines indicate the alternate position of parts of the object or the relative position of a missing part. They are composed of one long and two short evenly spaced dashes.

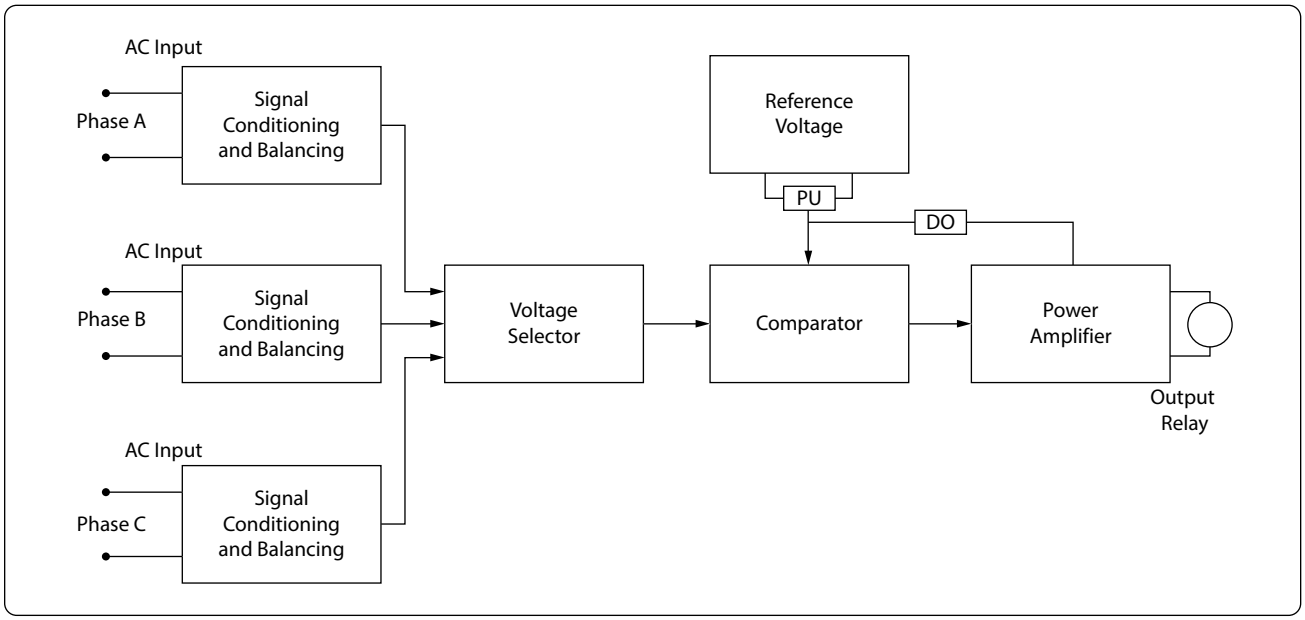


Figure 4-20. *Block diagram.*

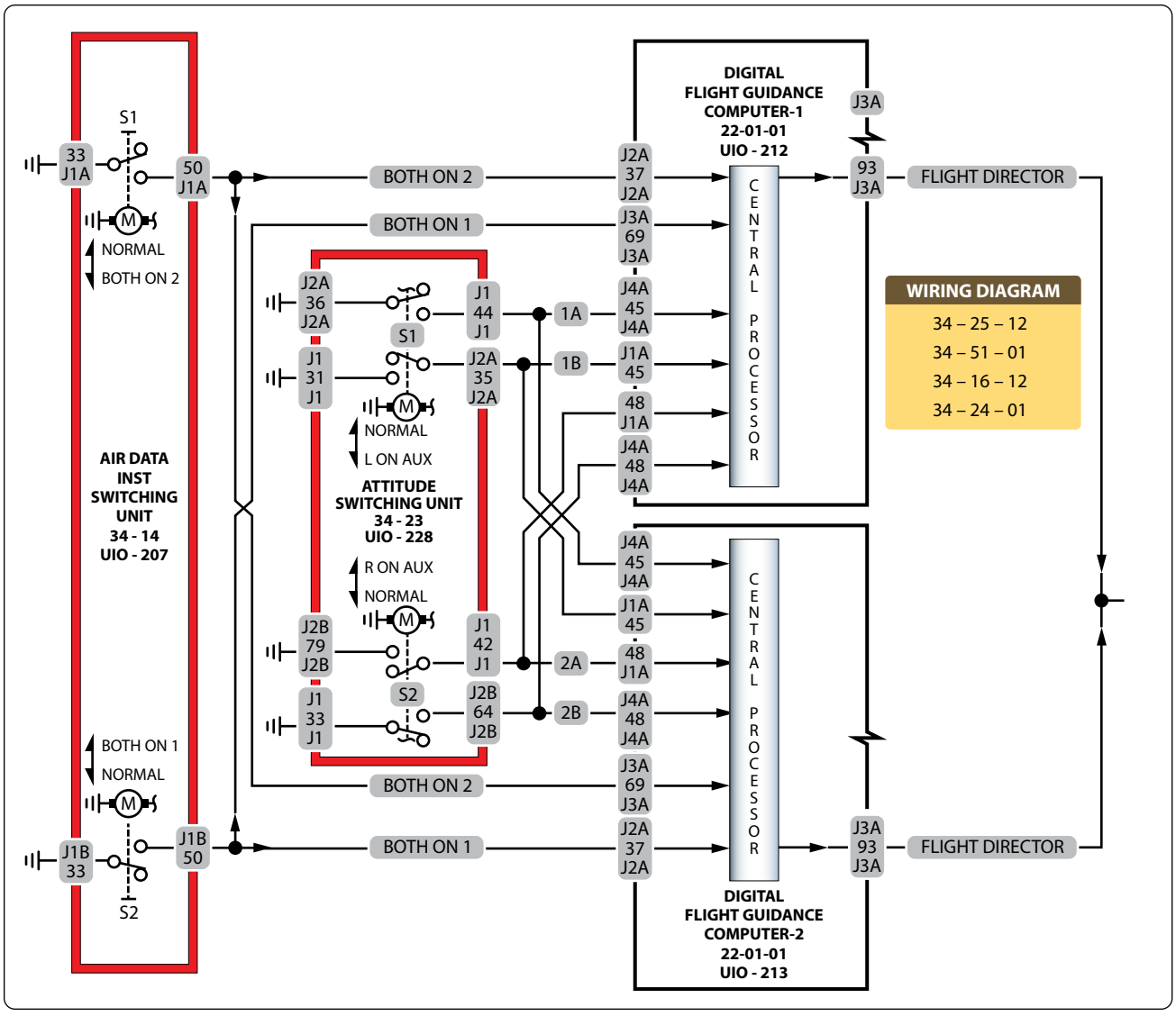


Figure 4-21. *Wiring diagram.*

Break Lines

Break lines indicate that a portion of the object is not shown on the drawing. Short breaks are made by solid, freehand lines. For long breaks, solid ruled lines with zigzags are used. Shafts, rods, tubes, and other such parts that have a portion of their length broken out have the ends of the break drawn as indicated in *Figure 4-25*.

Leader Lines

Leader lines are solid lines with one arrowhead. They indicate a part or portion that a note, number, or other reference applies.

Hidden Lines

Hidden lines indicate invisible edges or contours. Hidden lines consist of short dashes evenly spaced and are frequently referred to as dash lines.

Outline or Visible Lines

The outline or visible line is used for all lines on the drawing representing visible lines on the object. This is a medium-to-wide line that represents edges and surfaces that can be seen when the object is viewed directly.

Stitch Lines

Stitch lines are used to indicate the stitching or sewing lines on an article and consists of a series of very short dashes, approximately half the length of dash or hidden lines, evenly spaced. Long lines of stitching may be indicated by a series of stitch lines connected by phantom lines.

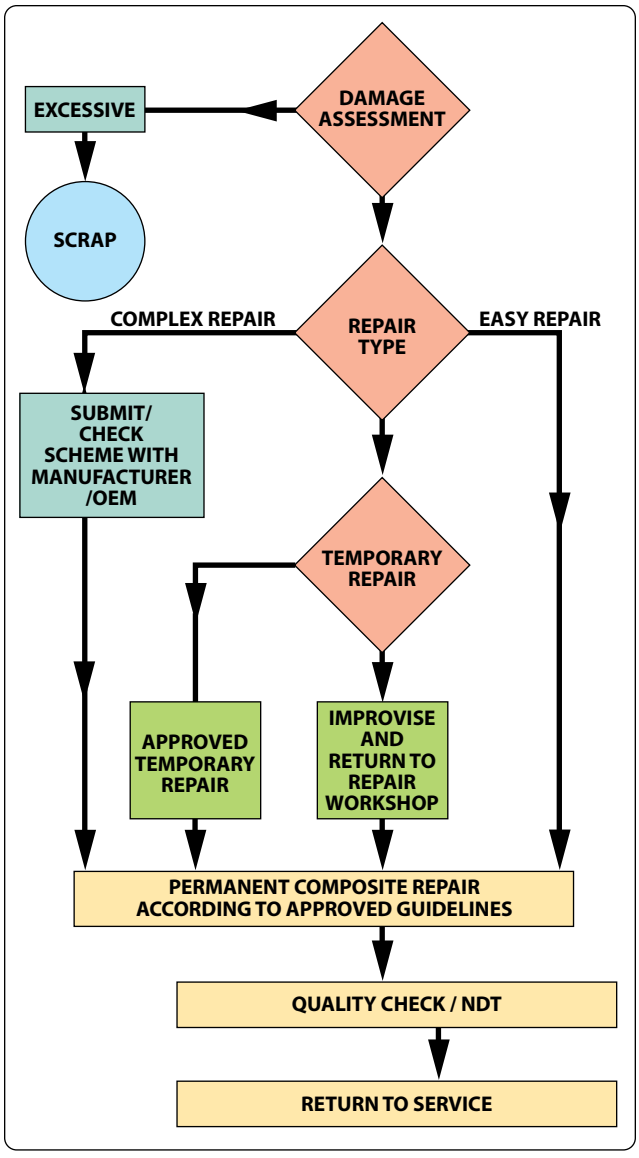


Figure 4-22. *Troubleshooting flowchart.*

Cutting Plane and Viewing Plane Lines

Cutting plane lines indicate the plane where a sectional view of the object is taken. In *Figure 4-25*, plane line A indicates the plane that section AA is taken. Viewing plane lines indicate the plane from where a surface is viewed.

Drawing Symbols

The drawings for a component are composed largely of symbols and conventions representing its shape and material. Symbols are the shorthand of drawing. They graphically portray the characteristics of a component with a minimal amount of drawing.

Material Symbols

Section line symbols show the kind of material from which the part is to be constructed. The material may not be indicated symbolically if its exact specification is shown elsewhere on the drawing. In this case, the more easily drawn symbol for cast iron is used for the sectioning, and the material specification is listed in the bill of materials or indicated in a note. *Figure 4-27* illustrates a few standard material symbols.

Shape Symbols

Symbols can be used to excellent advantage when needed to show the shape of an object. Typical shape symbols used on aircraft drawings are shown in *Figure 4-28*. Shape symbols are usually shown on a drawing as a revolved or removed section.

Electrical Symbols

Electrical symbols represent various electrical devices rather than an actual drawing of the units. *[Figure 4-29]* Having learned what the various symbols indicate, it becomes relatively simple to look at an electrical diagram and determine what each unit is, what function it serves, and how it is connected in the system.

Reading and Interpreting Drawings

Aircraft technicians do not necessarily need to be accomplished in making drawings. However, they must have a working knowledge of the information that is to be conveyed to them. They most frequently encounter drawings for construction and assembly of new aircraft and components, during modifications, and for making repairs.

A drawing cannot be read all at once any more than a whole page of print can be read at a glance. Both must be read one line at a time. To read a drawing effectively, follow a systematic procedure.

Upon opening a drawing, read the drawing number and the description of the article. Next, check the model affected, the latest change letter, and the next assembly listed. Having determined that the drawing is the correct one, proceed to read the illustration(s).

In reading a multiview drawing, first get a general idea of the shape of the object by scanning all the views. Then select one view for a more careful study. By referring back and forth to the adjacent view, it is possible to determine what each line represents.

Each line on a view represents a change in the direction of a surface, but another view must be consulted to determine what the change is. For example, a circle on one view may mean either a hole or a protruding boss, as in the top view of the object in *Figure 4-30*. Looking at the top view, we see two circles. However, the other view must be consulted to

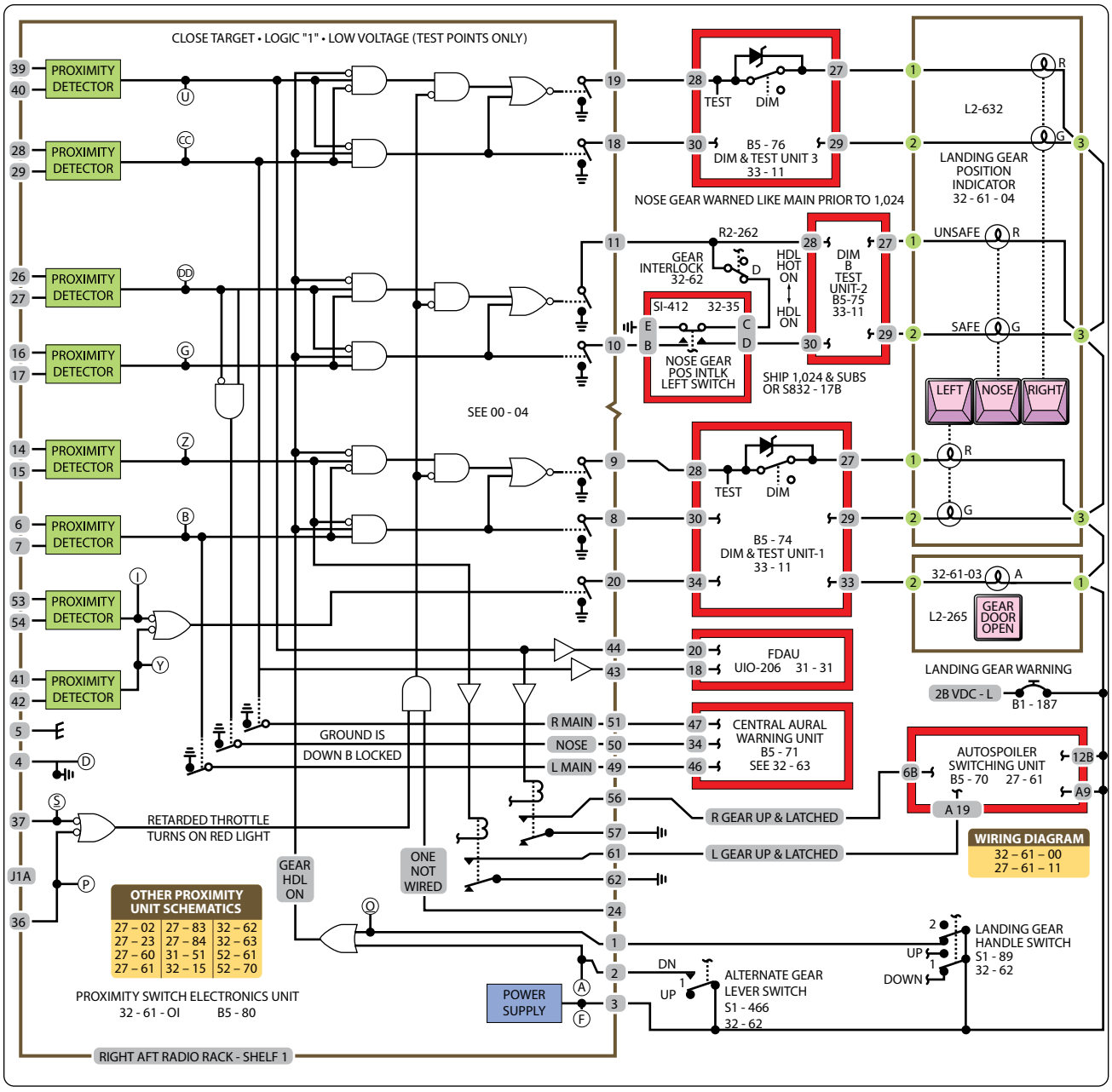


Figure 4-23. *Logic flowchart.*

determine what each circle represents.

A glance at the other view tells us that the smaller circle represents a hole, and the larger circle represents a protruding boss. In the same way, the top view must be consulted to determine the shape of the hole and the protruding boss.

It can be seen from this example that one cannot read a print by looking at a single view when more than one view is given. Two views do not always describe an object and when three views are given, all three must be consulted to be sure the shape has been read correctly.

After determining the shape of an object, determine its size. Information on dimensions and tolerances is given so that certain design requirements may be met. Dimensions are indicated by figures either with or without the inch mark. If no inch mark is used, the dimension is in inches. It is customary to give part dimensions and an overall dimension that gives the greatest length of the part. If the overall dimension is

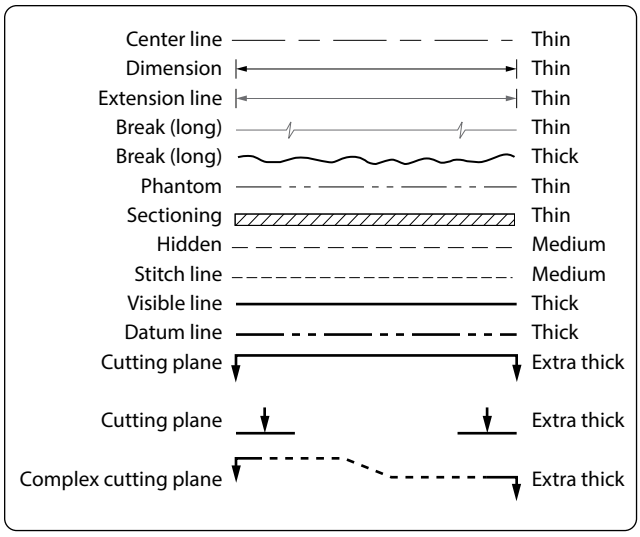


Figure 4-24. *The meaning of lines.*

missing, it can be determined by adding the separate part dimensions. Many drawings used for new aircraft and components are now using the metric system and millimeter (mm) is the unit used for these drawings.

Drawings may be dimensioned in decimals or fractions. This is especially true about tolerances. Instead of using plus and minus signs for tolerances, many figures give the complete dimension for both tolerances. For example, if a dimension is 2 inches with a plus or minus tolerance of 0.01, the drawing would show the total dimensions as:

2.01

1.99

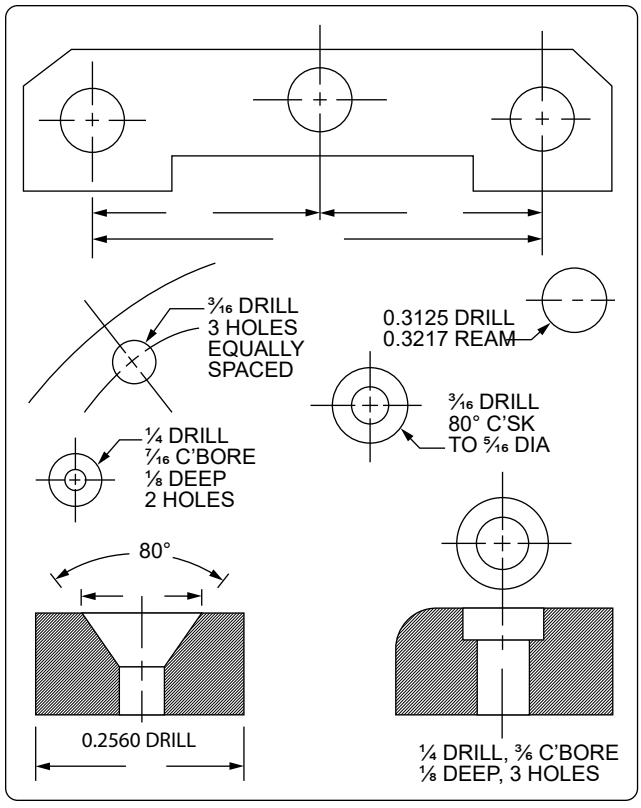


Figure 4-26. *Dimensioning holes.*

A print tolerance (usually found in the title block) is a general tolerance that can be applied to parts where the dimensions are noncritical. Where a tolerance is not shown on a dimension line, the print tolerance applies.

To complete the reading of a drawing, read the general notes and the content of the material block, find the various changes incorporated, and read the special information given in or

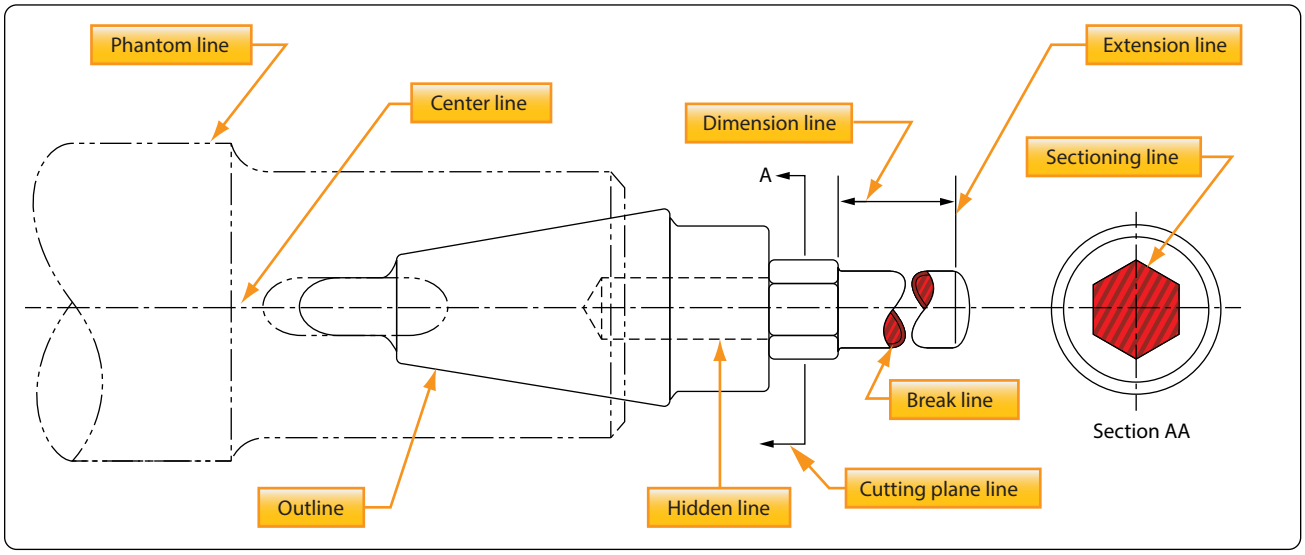


Figure 4-25. *Correct use of lines.*

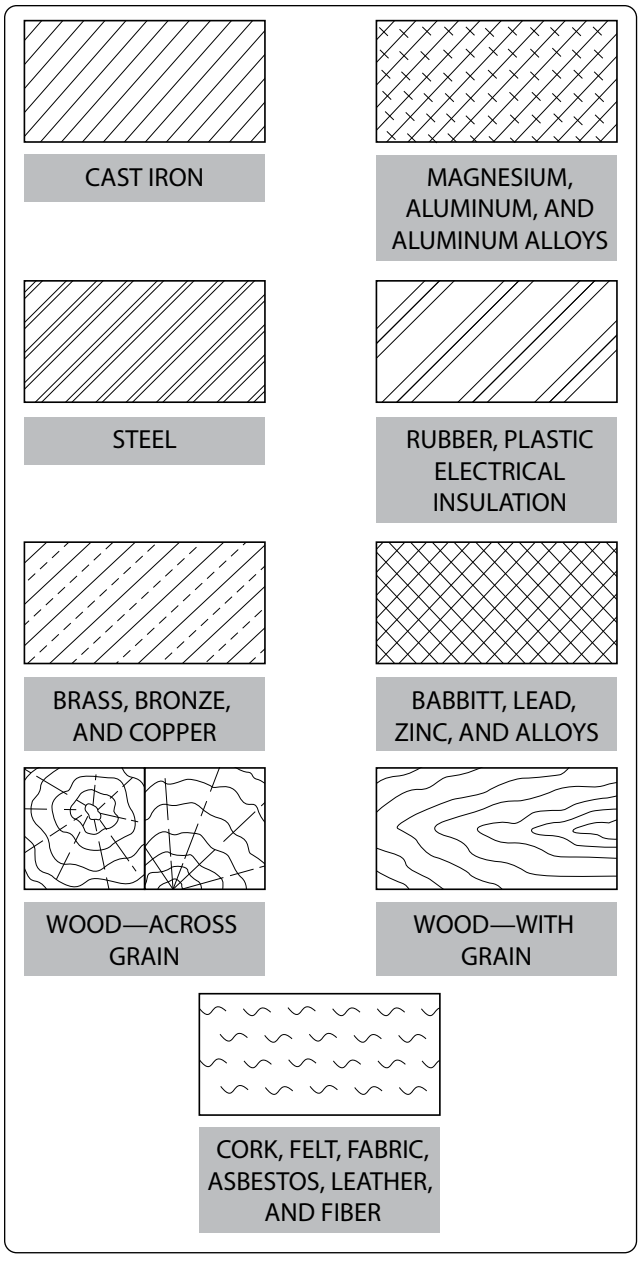


Figure 4-27. *Standard material symbols.*

near views and sections.

Drawing Sketches

A sketch is a simple rough drawing that is made rapidly and without much detail. Sketches may take many forms—from a simple pictorial presentation to a multi-view orthographic projection.

Just as aircraft technicians need not be highly skilled in creating drawings, they need not be accomplished artists. However, in many situations, they need to prepare a drawing to present an idea for a new design, a modification, or a repair method. The medium of sketching is an excellent way of accomplishing this.

The rules and conventional practices for making mechanical drawings are followed to the extent that all views needed to portray an object accurately are shown in their proper relationship. It is also necessary to observe the rules for correct line use and dimensioning. *[Figures 4-24* and *4-25]*

Sketching Techniques

To make a sketch, first determine what views are necessary to portray the object. Then block in the views using light construction lines. Next, complete the details, darken the object outline, and sketch extension and dimension lines. Complete the drawing by adding notes, dimensions, title, date, and when necessary, the sketcher's name. The steps in making a sketch of an object are illustrated in *Figure 4-31*.

Basic Shapes

Depending on the complexity of the sketch, basic shapes may be drawn in freehand or by use of templates. If the sketch is quite complicated or the technician is required to make frequent sketches, use of a variety of templates and other drafting tools is highly recommended.

Repair Sketches

A sketch is frequently drawn for repairs or for use in manufacturing a replacement part. Such a sketch must provide all necessary information to those who must make the repair or manufacture the part.

The degree that a sketch is complete depends on its intended use. Obviously, a sketch used only to represent an object pictorially need not be dimensioned. If a part is to be manufactured from the sketch, it should show all the necessary construction details.

Care of Drafting Instruments

Good drawing instruments are expensive precision tools. Reasonable care given to them during their use and storage can prolong their service life.

T-squares, triangles, and scales should not be used or placed where their surfaces or edges may be damaged. Use a drawing board only for its intended purpose and not in a manner that can mar the working surface.

Compasses, dividers, and pens provide better results with less annoyance, if they are correctly shaped and sharpened and are not damaged by careless handling. Store drawing instruments in a place where they are not likely to be damaged by contact with other tools or equipment. Protect compass and divider points by inserting them into a piece of soft rubber or similar material. Never store ink pens without first cleaning and drying them thoroughly.

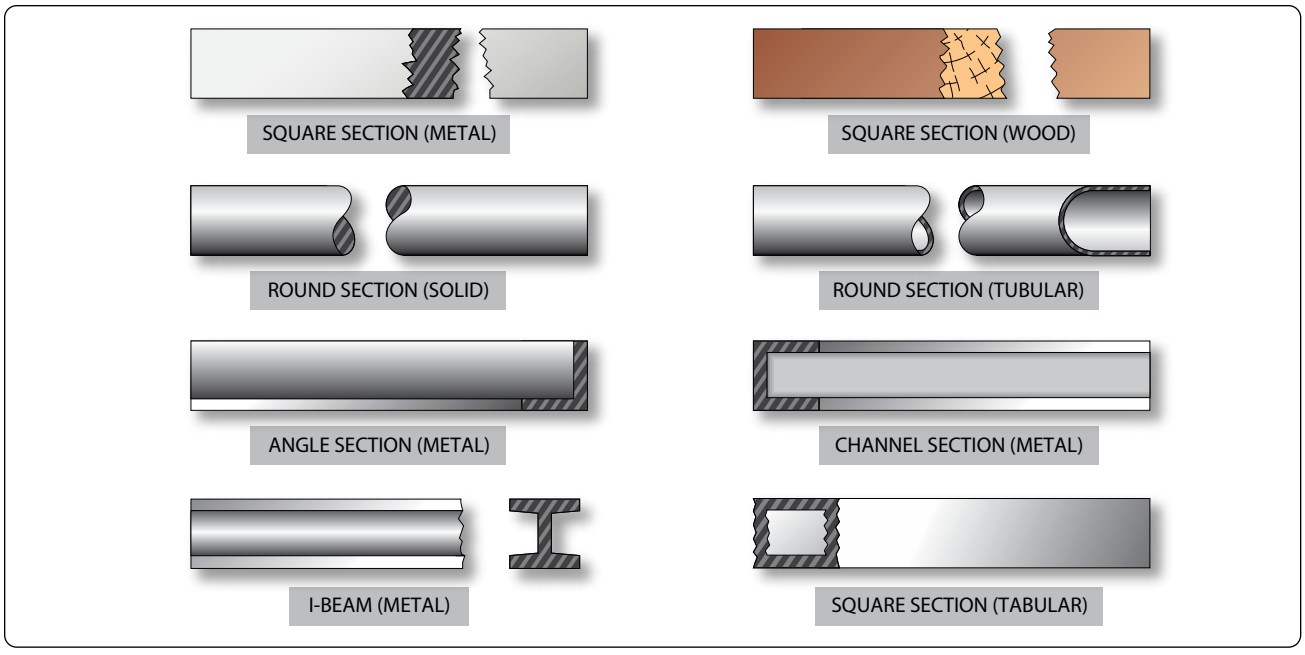


Figure 4-28. *Shape symbols.*

Graphs & Charts

Graphs and charts are frequently used to convey information graphically or information given certain conditions. They often utilize values shown on the "x" and "y" axes that can be projected up and across to arrive at a specific result. Also, when data is entered into a computer database, software programs can create a variety of different bar graphs, pie charts, and so forth, to graphically represent that data.

Reading & Interpreting Graphs & Charts

When interpreting information shown on graphs and charts, it is extremely important that all the notes and legend information be carefully understood to eliminate any misinterpretation of the information presented.

Nomograms

A nomogram is a graph that usually consists of three sets of data. Knowledge of any two sets of data enables the interpreter to obtain the value for the third unknown corresponding value. One type of nomogram consists of three parallel scales graduated for different variables, so that when a straight edge connects any two values, the third can be read directly. Other types may use values on the "x" and "y" axes of a graph with the third corresponding value determined by the intersection of the "x" and "y" values with one of a series of curved lines. *Figure 4-32* is an example of a nomogram that shows the relationship between aviation fuels, specific weight, and temperature.

Microfilm & Microfiche

The practice of recording drawings, parts catalogs, and maintenance and overhaul manuals on microfilms was utilized extensively in the past. Microfilm is available as regular 16 mm or 35 mm film. Since 35 mm film is larger, it provides a better reproduction of drawings. Microfiche is a card with pages laid out in a grid format. Microfilm and microfiche require use of special devices for both reading and printing the information.

Most modern aircraft manufacturers have replaced microfilm and microfiche with digital storage methods utilizing CDs, DVDs, and other data storage devices. A great deal of service and repair information for older aircraft has been transferred to digital storage devices. However, there may still be a need to access information using the old methods. A well-equipped shop should have available, both the old microfilm and microfiche equipment, as well as new computer equipment.

Digital Images

Though not a drawing, a digital image created by a digital camera can be extremely helpful to aviation maintenance technicians in evaluating and sharing information concerning the airworthiness or other information about aircraft. Digital images can be rapidly transmitted over the World Wide Web as attachments to e-mail messages. Images of structural fatigue cracks, failed parts, or other flaws, as well as desired design and paint schemes, are just a few examples of the types of digital images that might be shared by any number of users over the Internet. *Figure 4-33* is a digital image of damage to a composite structure taken with a simple digital camera.

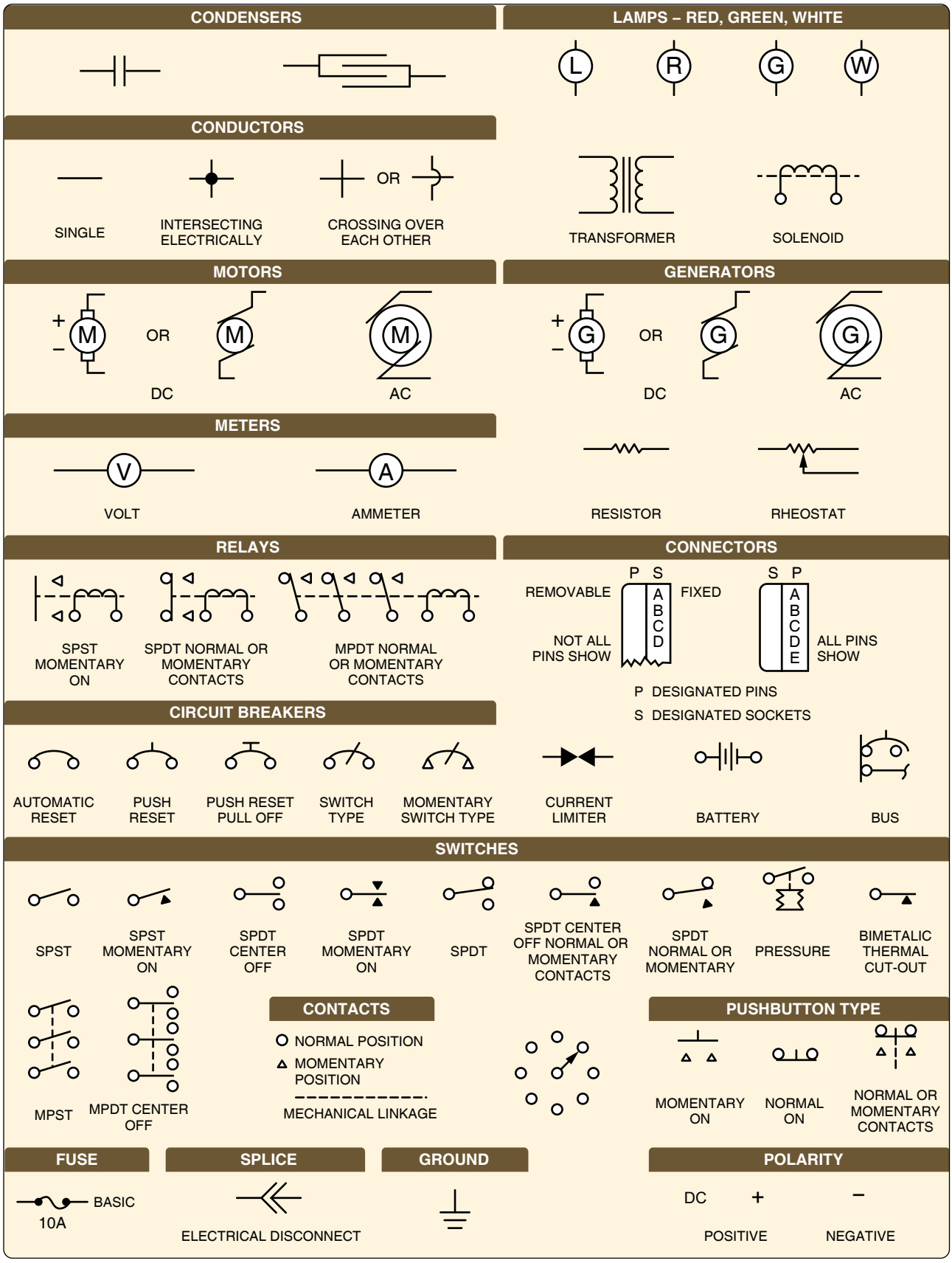


Figure 4-29. *Electrical symbols.*

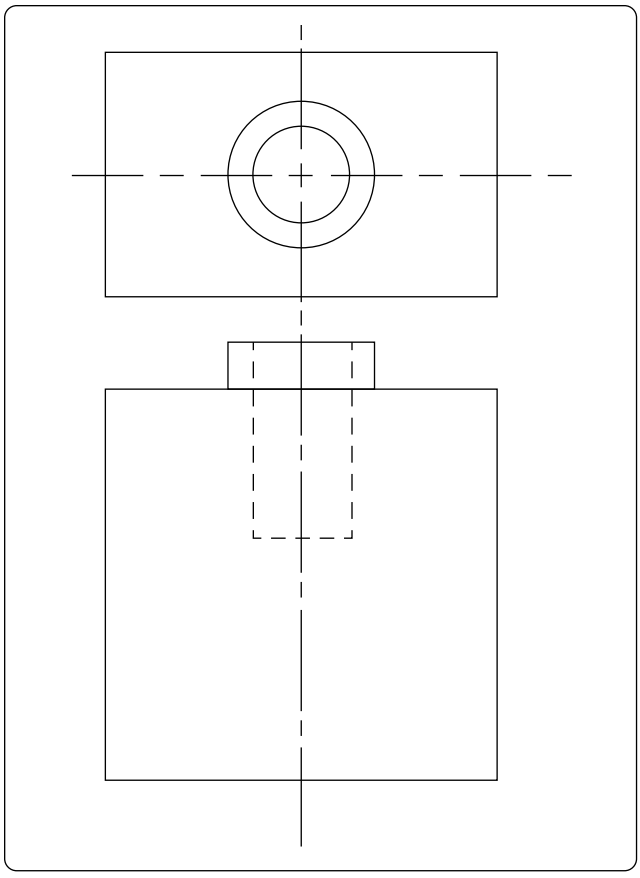


Figure 4-30. *Reading views.*

To provide information about the extent of the damage, a measurement scale, or other object, such as a coin, can be placed near the area of concern before the picture is taken. Also, within the text of the e-mail, the technician should state the exact location of the damage, referenced to fuselage station, wing station, and so forth.

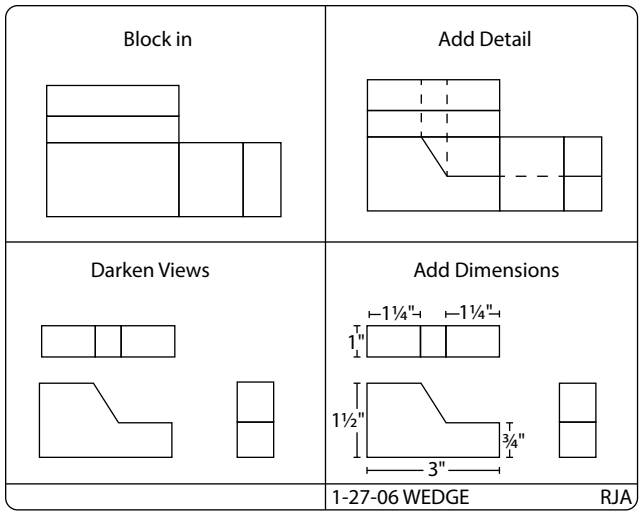


Figure 4-31. *Steps in sketching.*

Density Variation of Aviation Fuel
Based on Average Specific Gravity

Fuel	Average Specific Gravity at 15 °C (59 °F)
Aviation Kerosene Jet A and Jet A1	0.812
Jet B (JP-4)	0.785
AV Gas Grade 100/130	0.703

NOTE: The fuel quantity indicator is calibrated for correct indication when using Aviation Kerosene Jet A and Jet A1. When using other fuels, multiply the indicated fuel quantity in pounds by 0.99 for Jet B (JP-4) or by 0.98 for Aviation Gasoline (100/130) to obtain actual fuel quantity in pounds.

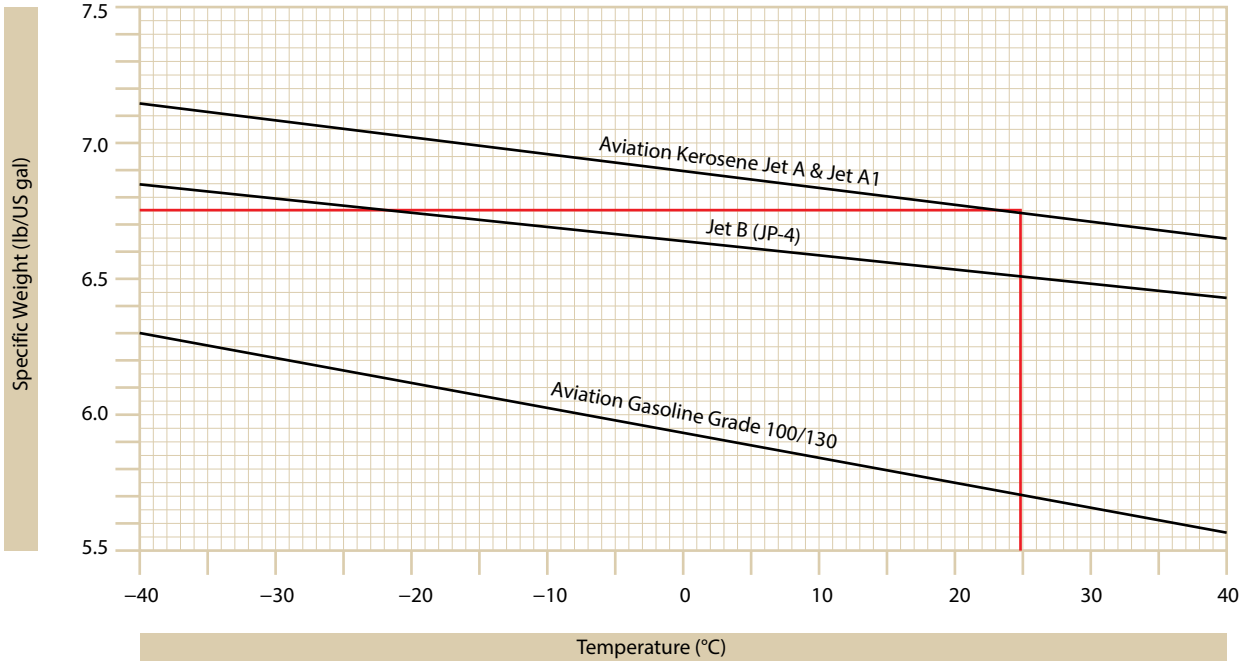


Figure 4-32. *Nomogram.*



Figure 4-33. *Digital image of damage.*